

Introduction

Scientific Machine Learning and Simulation Intelligence

Who are we?

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Who are you?

Scientific Background?

Experience in Machine Learning ?

Programming Experience? Python?

What is this course about?

Simulation Intelligence

Topics:

1. Introduction to Differential Equations and Differential programming
2. Physics-Infused Surrogate Modelling
 - Neural Ordinary Differential Equations (Neural ODE)
 - Reduced-Order Models (ROMs)
 - Physics-Informed Neural Networks (PINNs)
 - Neural Operators (NO)
3. Simulation-Based Inference
4. Uncertainty Estimation
5. Causal Reasoning

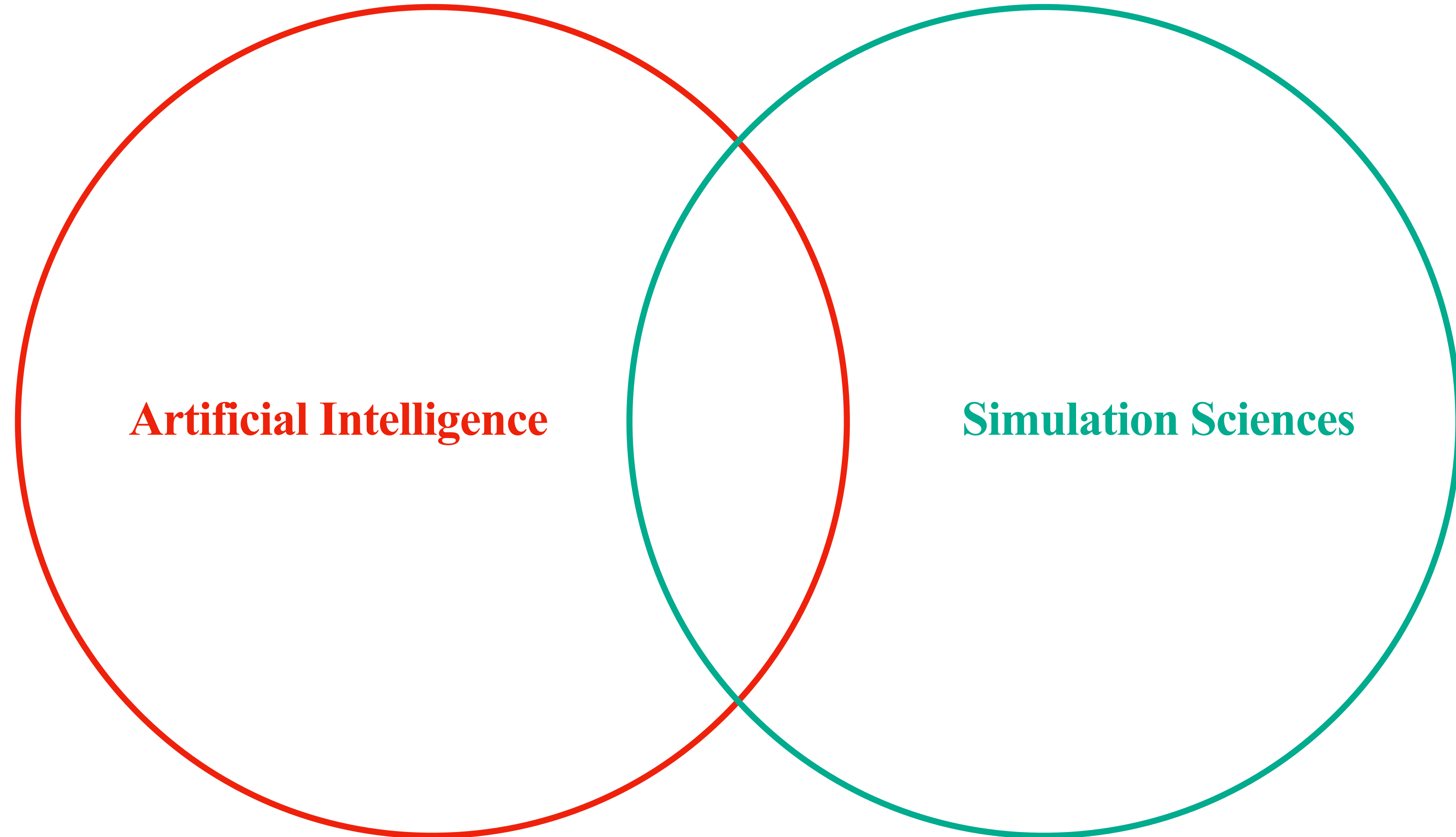
Material

Textbook and online material

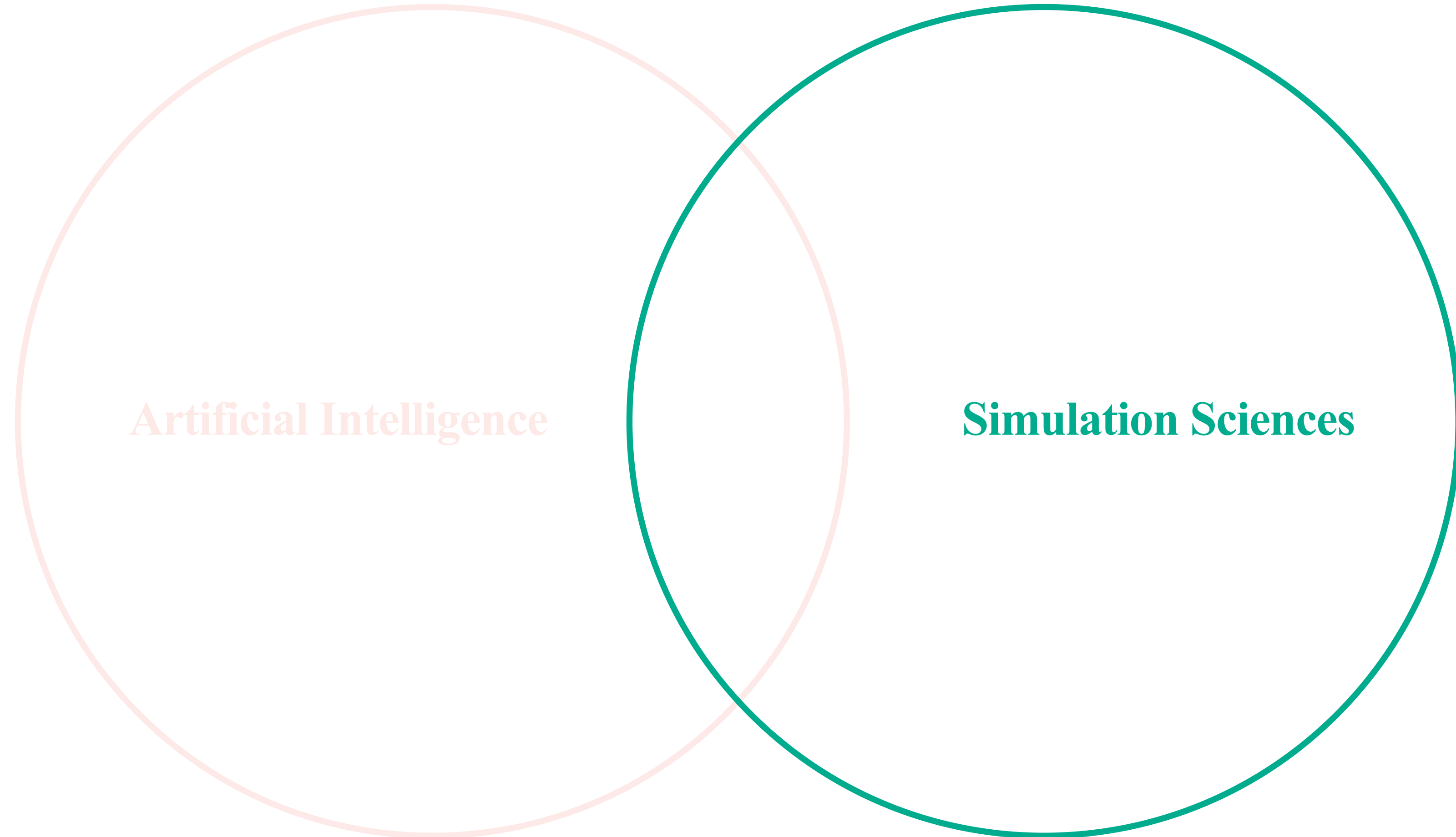
Exam

Scientific project on one of the topics

Simulation intelligence



Simulation intelligence

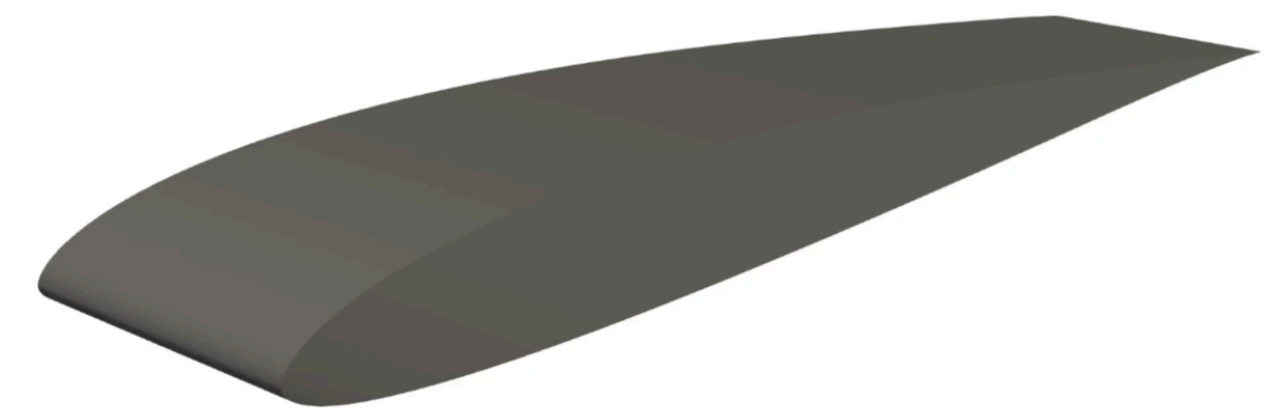


Simulation Sciences: Differential Equations

In fluid dynamics, **physics-based** models are used to predict how fluids move and interact with their surroundings, from the air flowing around an aircraft to the circulation of the oceans. The outcomes guide the design of vehicles and buildings, weather forecasting, and the assessment of environmental risks.

Example: In aerodynamics, models estimate the lift and drag acting on a wing or a vehicle. The results feed directly into design decisions, such as reducing fuel consumption or ensuring the safety of a bridge under wind loads.

These models rest on fundamental physical laws, namely the conservation of mass, momentum and energy. Combined, they give rise to the **Navier-Stokes equations**, a system of PDEs that in principle describes virtually every flow we encounter, and that embeds decades of theory, experiments and observations.



Simulation Sciences: Differential Equations

A physics-based law is a mathematical statement derived from fundamental principles of physics, such as the conservation of mass, momentum and energy, and is intended to be universally true: it holds regardless of the specific geometry, scale or conditions of the problem. Its purpose is to describe the *why* and the *how* of a process from the ground up, rather than simply reproducing what is observed. Instead of asking "what happened in this case?", it asks "what mechanism governs this class of phenomena?", which is what makes it predictive even in situations that have never been measured.

The typical form of these laws is a **differential equation**. This is a natural consequence of how conservation works: the balance of a quantity (mass, momentum, energy) within an infinitesimal control volume relates its rate of change in time to what flows through the boundaries and to what is produced or removed inside. Taking the volume to zero size yields a relation between a field and its derivatives in space and time.

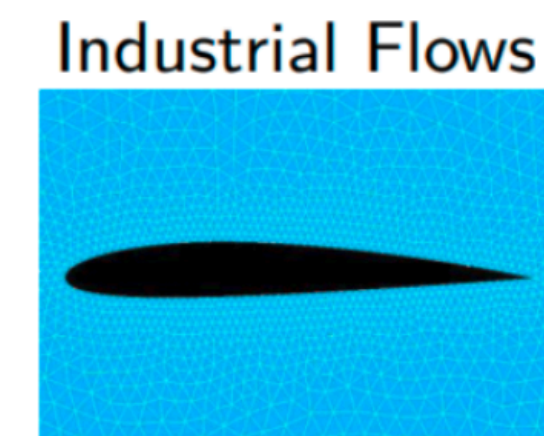
$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_j) = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_i u_j) = - \frac{\partial p}{\partial x_i}$$

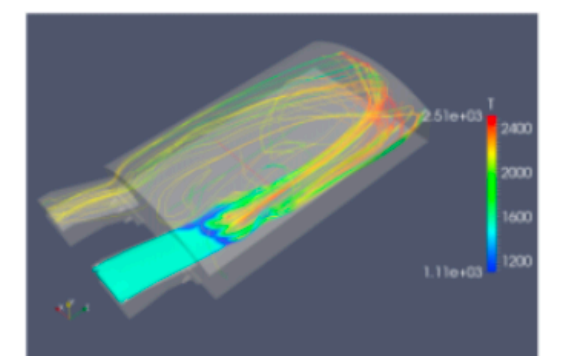
$$\frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_j} ((\rho E + p) u_j) = 0$$



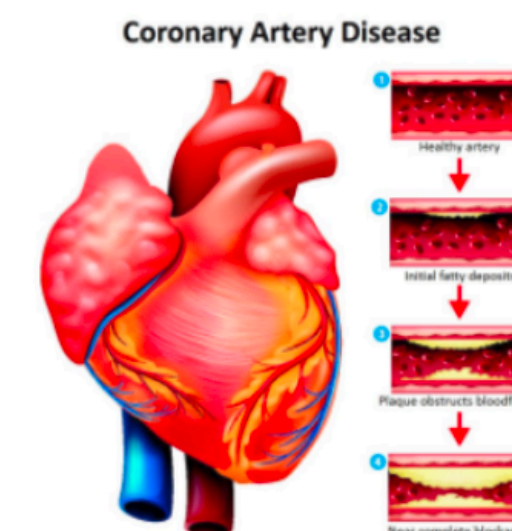
Naval Eng.



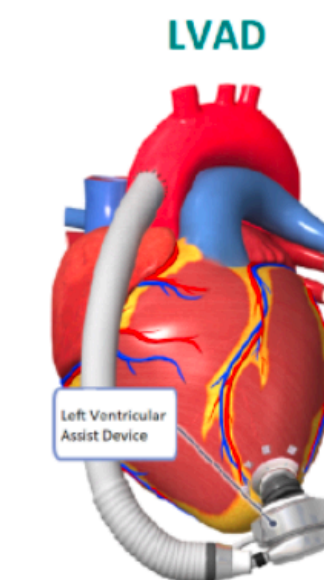
Aeronautics



Industrial App.

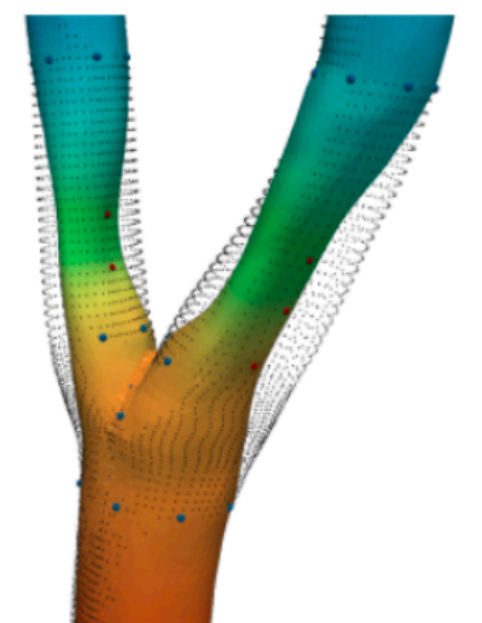


Coronary Artery Disease



LVAD

Biomedical Applications



Simulation Sciences: Differential Equations

Equations involving derivatives, representing how things change in space and time. Used to model dynamical systems in science, engineering, finance and more.

Types of Differential Equations:

- A. Ordinary Differential Equations (ODEs)
- B. Partial Differential Equations (PDEs)

Simulation Sciences: ODEs

An equation involving derivative with respect to only one independent variable (usually time):

$$\frac{dy}{dt} = f(y, t)$$

Example: Newton's Law of Cooling: $\frac{dT}{dt} = -k(T - T_{env})$

Where T is the temperature.

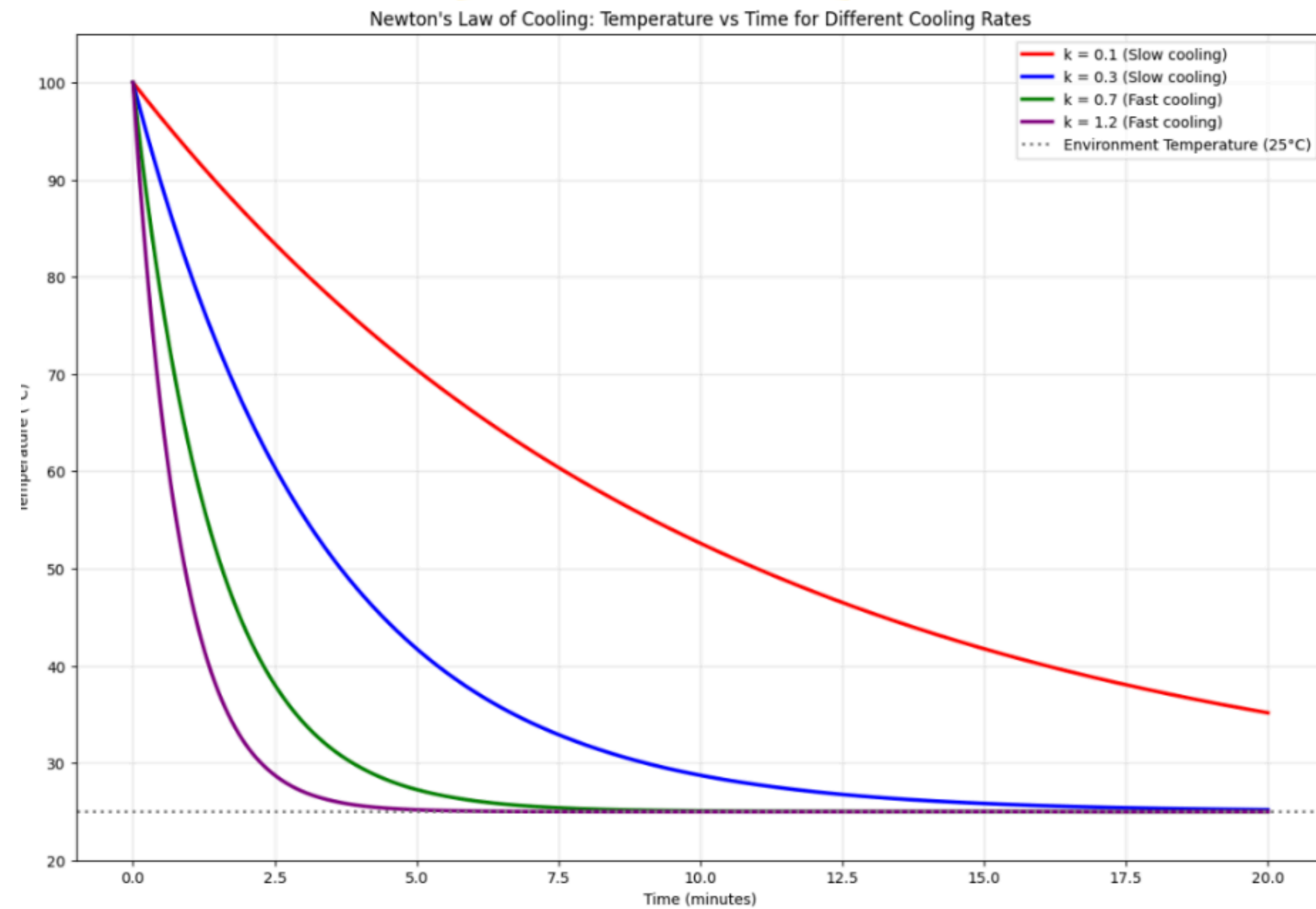
Initial conditions? Vector field?

Conditions at other times? Hybrid ODEs?

Applications: modeling population growth, pendulum motion and electrical circuits

Simulation Sciences: ODEs

Example: Newton's Law of Cooling:



Simulation Sciences: Differential equations

How to solve differential equations to understand how systems evolve?

You can either consider:

1. **Exact Methods:** provide precise analytical solutions
2. **Numerical Methods:** approximate solutions when exact methods are impractical or not available at all

Simulation Sciences: ODEs

Analytical methods to find the solution of differential equations:

Common methods:

1. **Separation of Variables:** Used when variables can be isolated on either side of the equation
2. **Integrating Factor:** Common for linear first-order ODEs
3. **Characteristic Equation:** Often used for linear, constant-coefficient ODEs
4. **Fourier and Laplace Transform:** Useful for PDEs and complex boundary conditions

Limitations: Only works for certain types of equations, can be challenging for complex systems

Example: solve $\frac{dy}{dx} = ky$

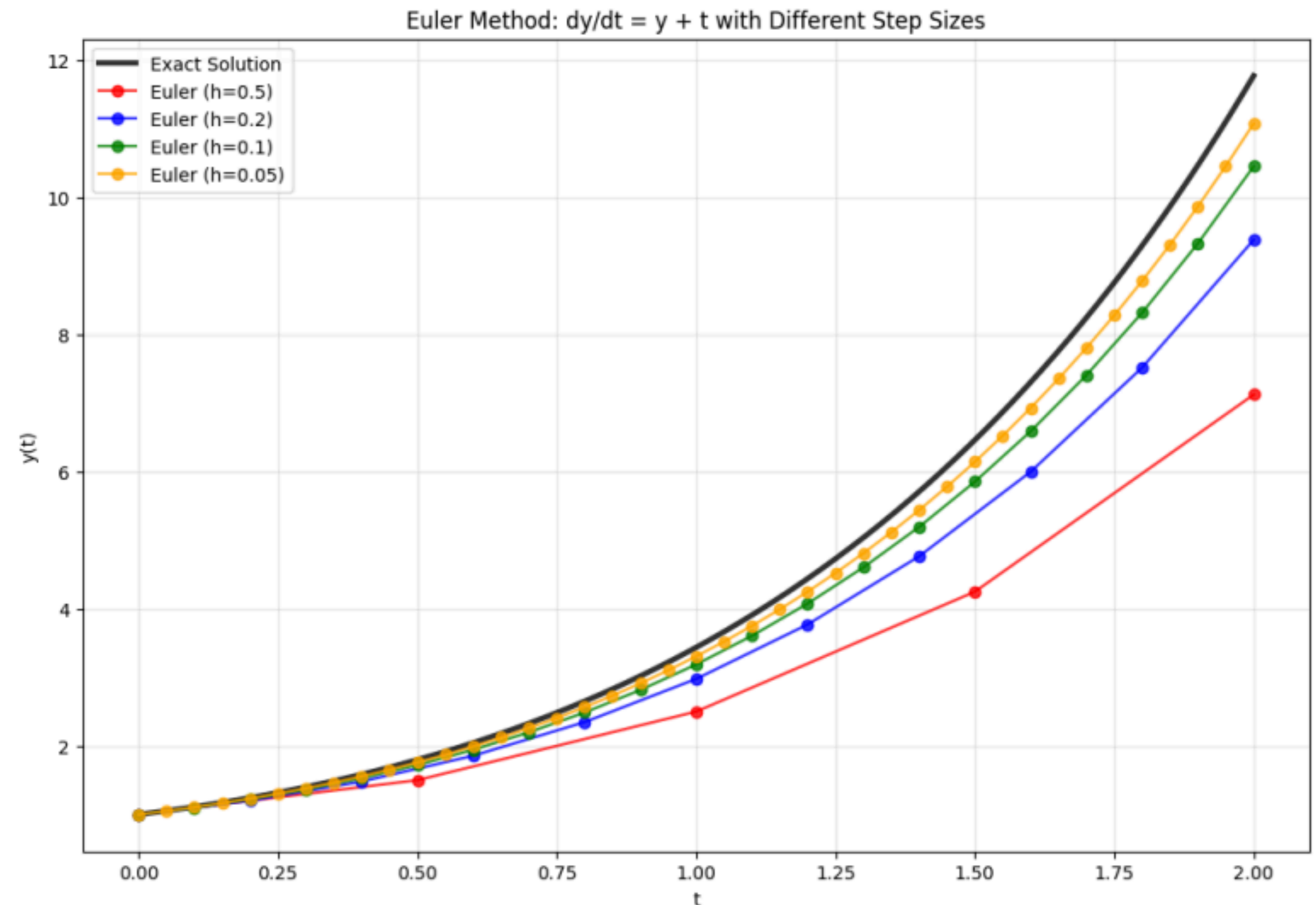
Solution $y = Ce^{kx}$

Applications: Population growth, radioactive decay, simple financial models

Simulation Sciences: ODEs

Solve $\frac{dy}{dt} = y + t$ using Euler's Method.

Choose a starting point $y(0) = 1$, calculate the next y -value using $y_{n+1} = y_n + hf(t_n, y_n)$



[Notebook on ODEs](#)

Simulation Sciences: ODEs

Solve $\frac{dy}{dt} = y + t$ using Runge Kutta's Method.

The RK4 method uses four intermediate calculations:

$$k_1 = h \cdot f(t_n, y_n)$$

$$k_2 = h \cdot f(t_n + h/2, y_n + k_1/2)$$

$$k_3 = h \cdot f(t_n + h/2, y_n + k_2/2)$$

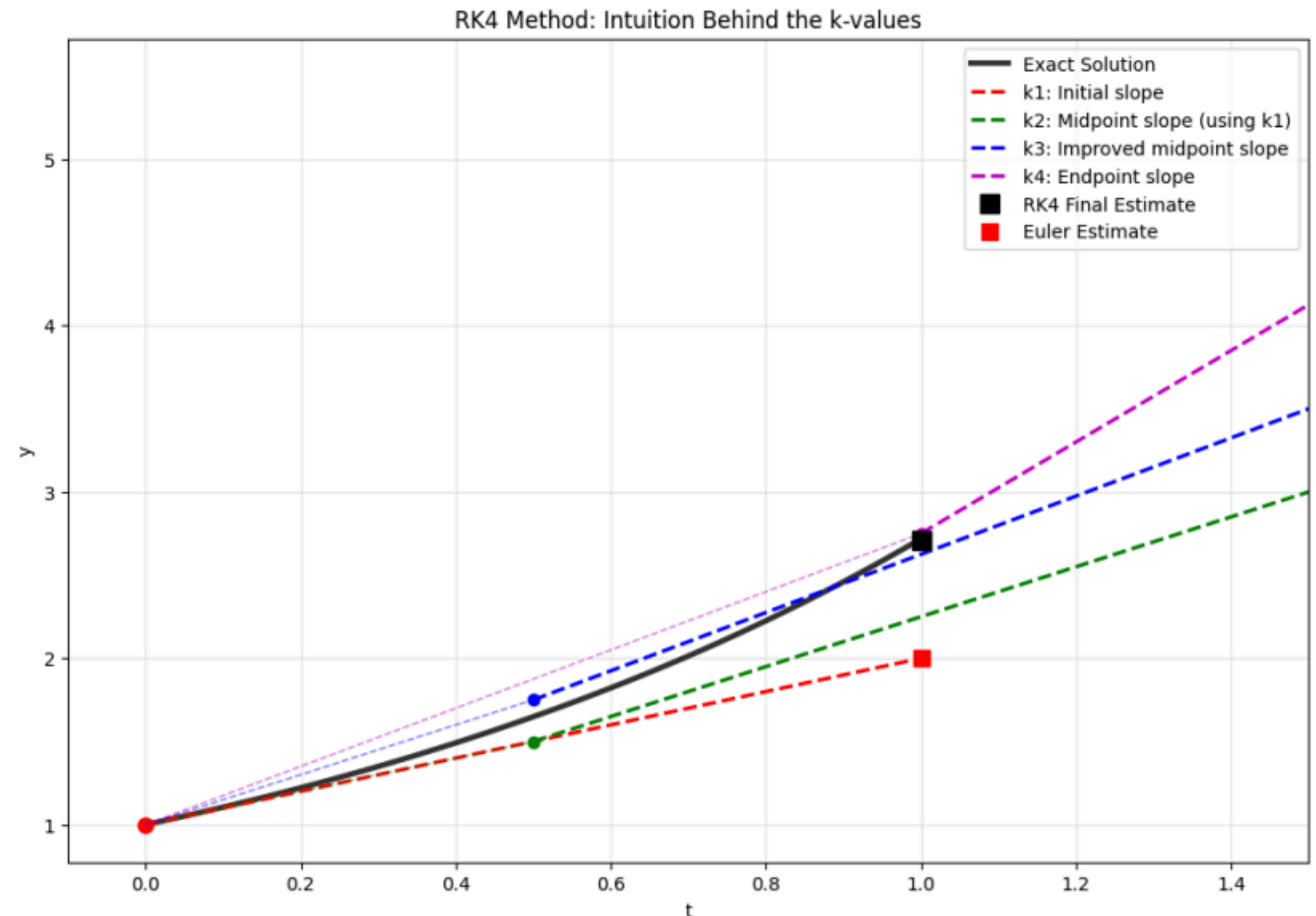
$$k_4 = h \cdot f(t_n + h, y_n + k_3)$$

Then update:

$$y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)/6$$

$$t_{n+1} = t_n + h$$

[Notebook on ODEs](#)



Simulation Sciences: ODEs

Solve $\frac{dy}{dt} = y + t$ using Runge Kutta's Method.

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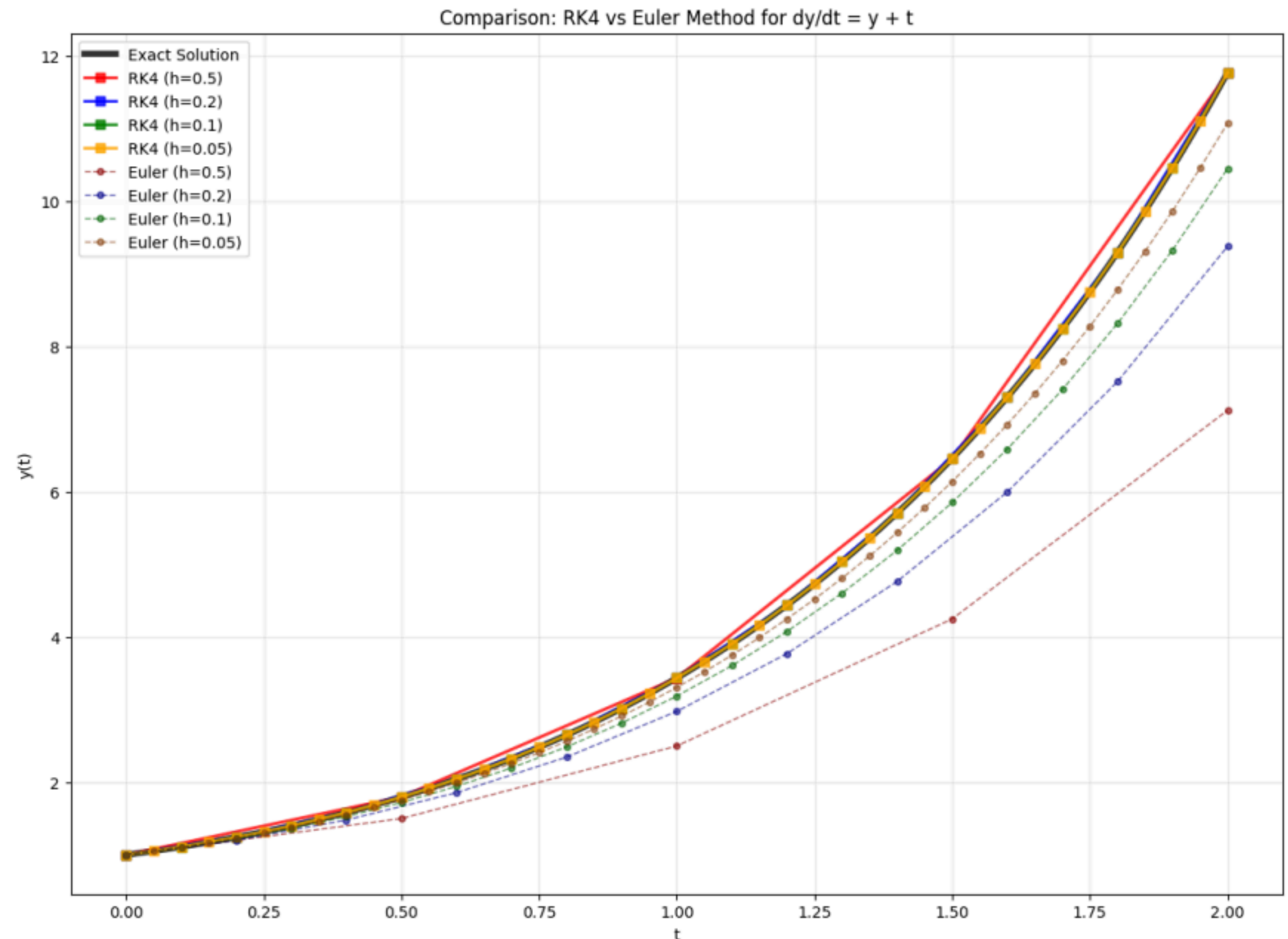
$$k_4 = h \cdot f(t_n + h, y_n + k_3)$$

Then update:

$$y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)/6$$

$$t_{n+1} = t_n + h$$

[Notebook on ODEs](#)



Simulation Sciences: ODEs

Approximate the solution by calculating values at discrete points

Common methods:

- Euler's Method: simple approach, uses step-by-step approximation
- Runge-Kutta Methods: more accurate than Euler's, with high-order precision
- Finite Difference Methods: approximate derivative, commonly used for PDEs

Advantage: works for complex equations and systems without exact solutions

Limitations: only an approximation, with possible errors based on the discretisation

Applications: engineering simulations, financial forecasting, physics simulation

Example: solve $\frac{dy}{dt} = y + t$ starting from point $y(0) = y_0$ using different methods

Simulation Sciences: PDEs

Partial differential equations involve derivatives not only in time but also spatially. Of course such form is necessary when you expect that the dynamics of your physical phenomena will depend based on the local characteristics.

$$\frac{\partial u}{\partial t} = f\left(u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}\right)$$

A classical example is thermal diffusion which can be approximated by means of the heat equation:

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

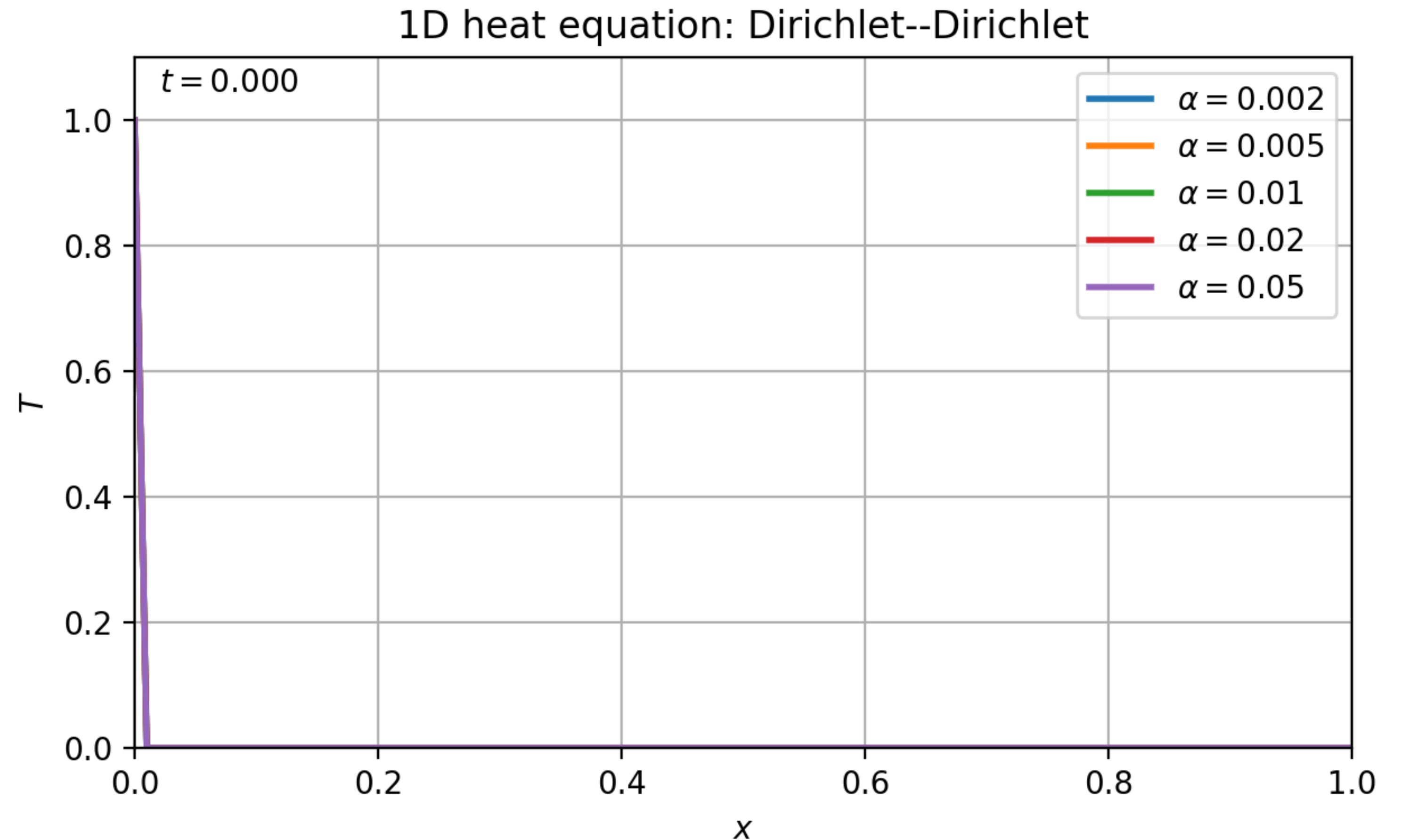
Depending on the different initial/boundary conditions I could get a different behaviour in the spatial domain. Also, I might even have a thermal diffusivity which depends itself on space and so:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(\kappa(x) \frac{\partial u}{\partial x} \right)$$

Simulation Sciences: PDEs

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

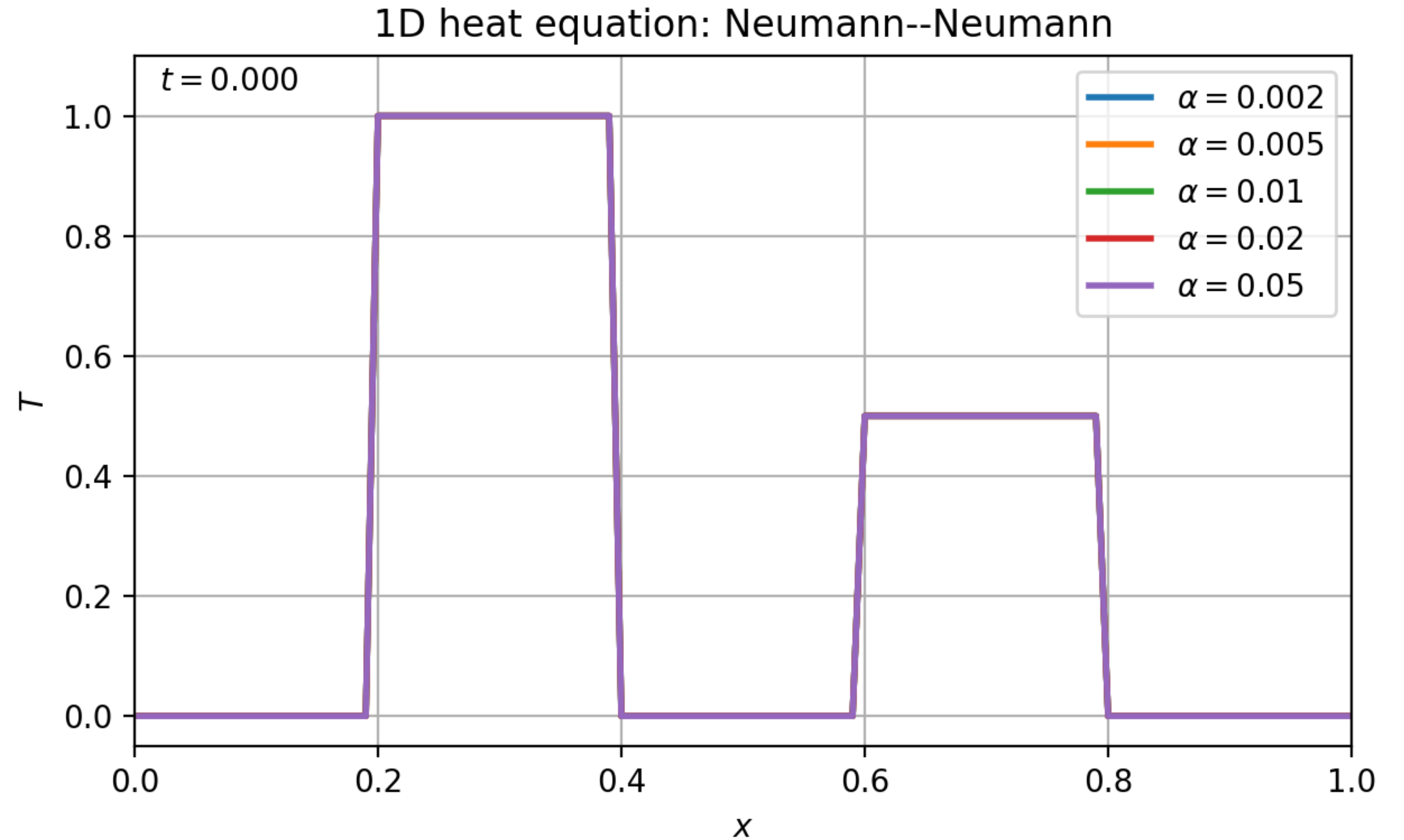
This is a classically realistic scenario: you have an **hot** condition on the left (your heater) and on the other side you have you **cold** little room. Depending on the how strongly heat propagates into the room the faster the room will get warm



Simulation Sciences: PDEs

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

Or maybe you have a series of localised heat sources and you want to see how fast heat will flow in the room as soon as you turn everything off.



Simulation Sciences: PDEs

A partial differential equation relates an unknown function of several independent variables, for instance space and time, to its partial derivatives. The **order** of a PDE is the order of the highest derivative appearing in it, and it is the first property we use to characterise an equation.

$$F \left(\mathbf{x}, t, u, \frac{\partial u}{\partial x_i}, \frac{\partial u}{\partial t}, \frac{\partial^2 u}{\partial x_i \partial x_j}, \dots \right) = 0$$

First-order equations typically describe transport phenomena, while second-order equations are the most common in physics, since they arise from balances involving fluxes such as diffusion and viscous stress. Higher orders are rarer and appear, for instance, in beam bending or in models of pattern formation.

First-order transport equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

Second-order heat equation

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

Second-order wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Simulation Sciences: PDEs

Depending on their order and type, PDEs have different qualitative properties, which tell us a lot about the solution before we compute anything. First-order equations describe transport: information travels **along characteristics at finite speed**, and the initial profile is carried along (possibly steepening into shocks if nonlinear). Second-order equations depend on their type. Elliptic and parabolic ones, such as Poisson and heat, satisfy a maximum principle, so the solution remains bounded by the boundary and initial data, and diffusion smooths out irregularities at **infinite speed**. Hyperbolic ones, such as the wave equation, propagate information at finite speed and do not smooth the solution, so discontinuities persist.

These properties determine which conditions must be imposed, how the equation should be discretised, and where a neural network is likely to struggle: sharp fronts and shocks are much harder to learn than smooth diffusive solutions.

First-order transport equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

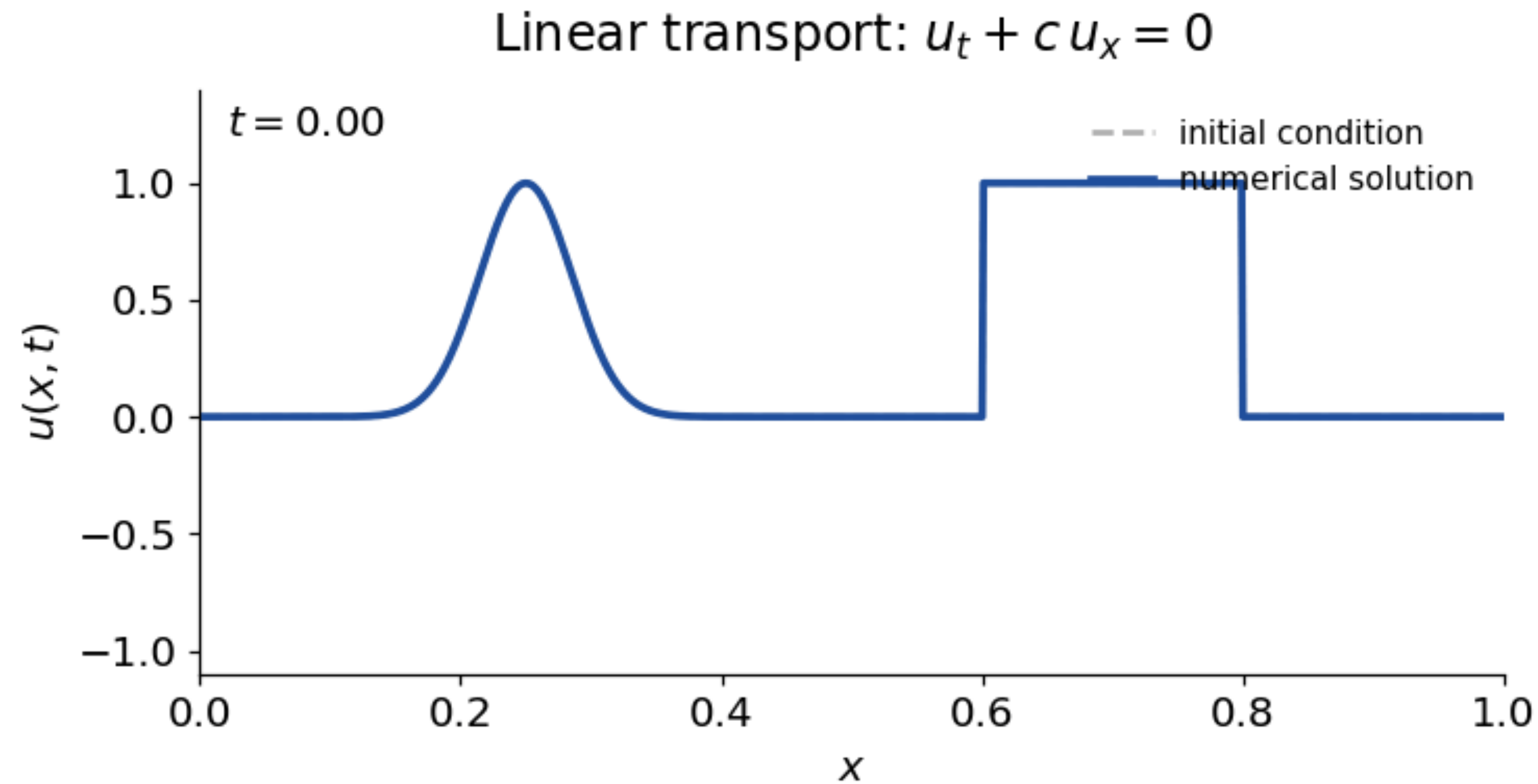
Second-order heat equation

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

Second-order wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

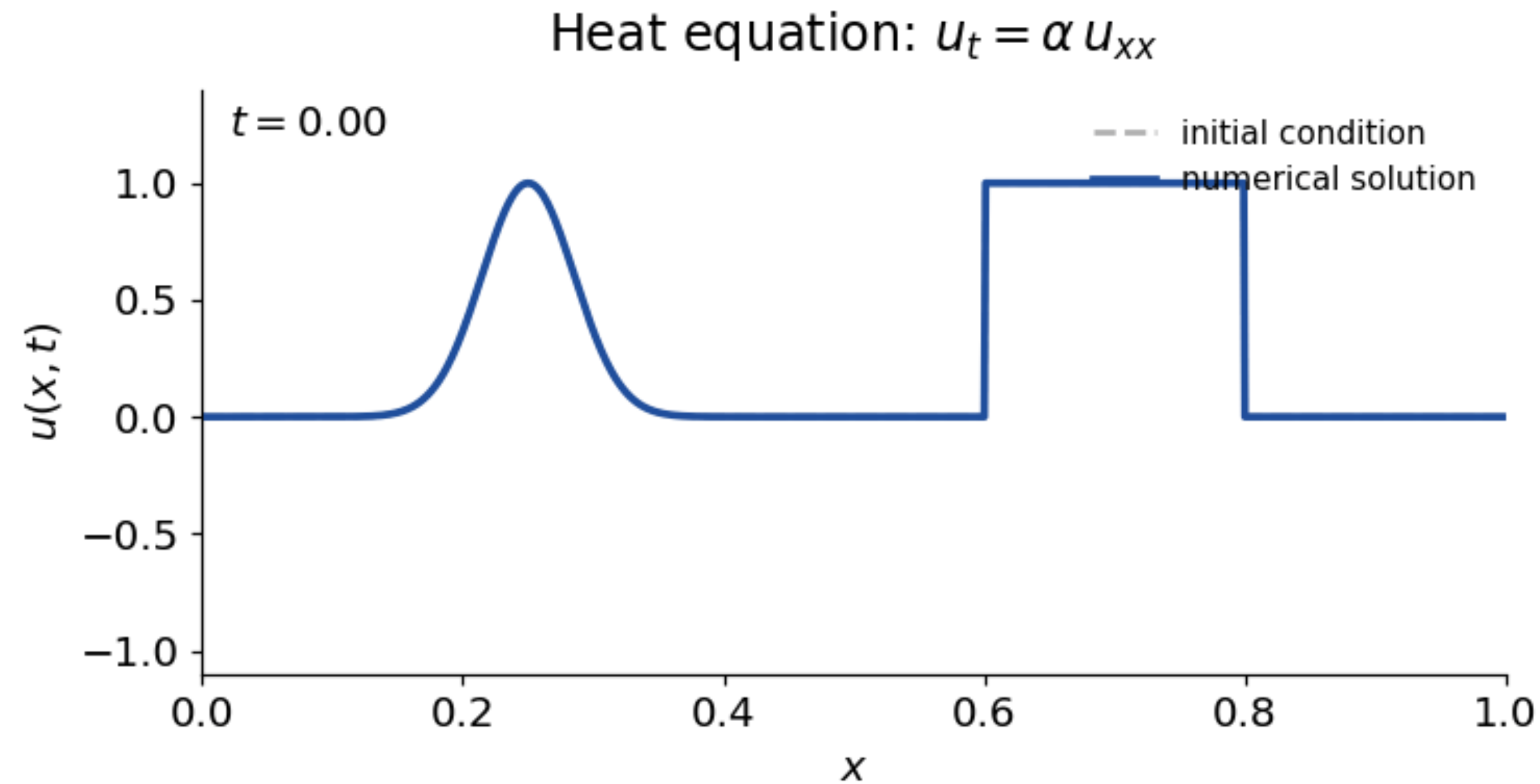
Simulation Sciences: PDEs



First-order transport equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

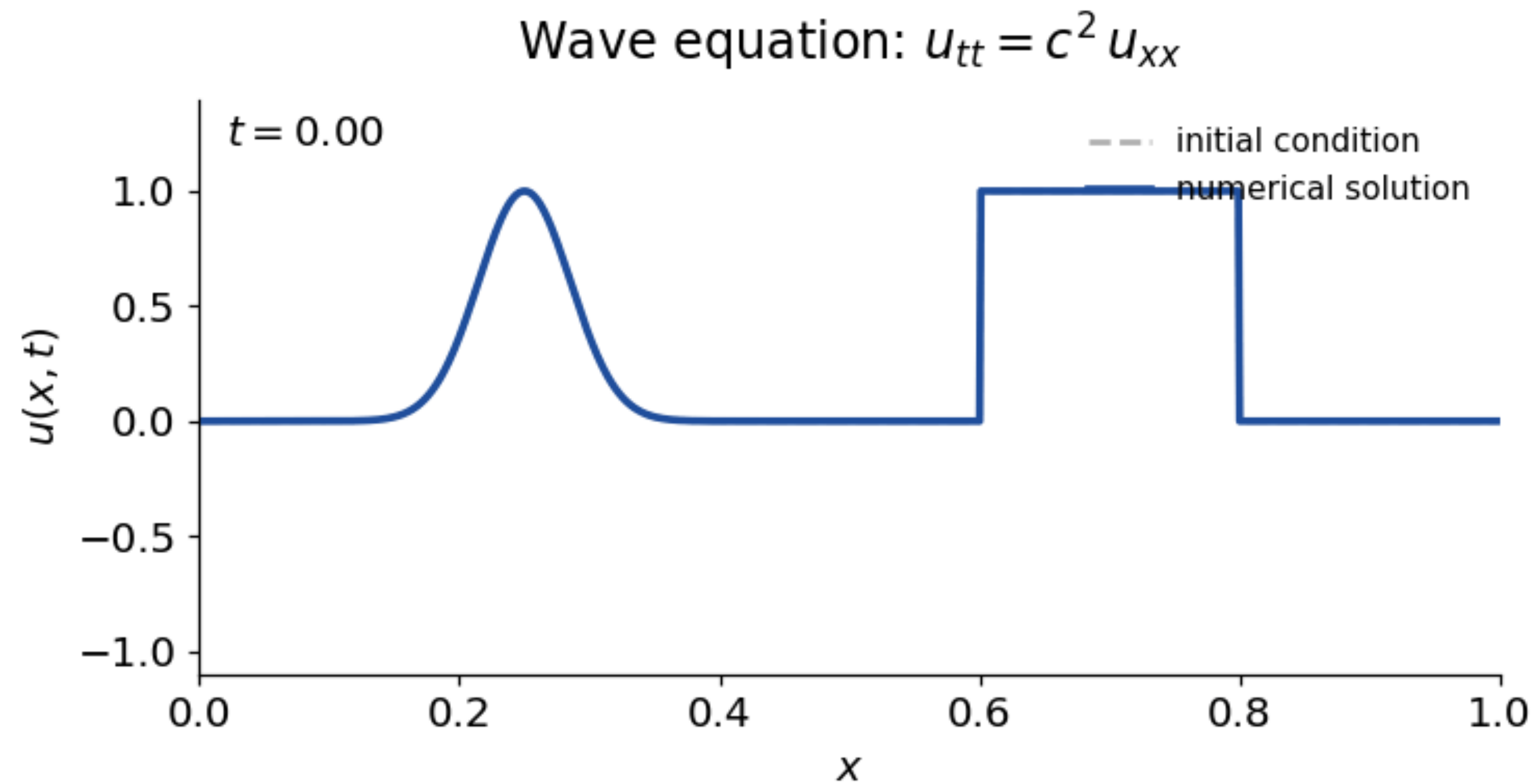
Simulation Sciences: PDEs



Second-order heat equation

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

Simulation Sciences: PDEs



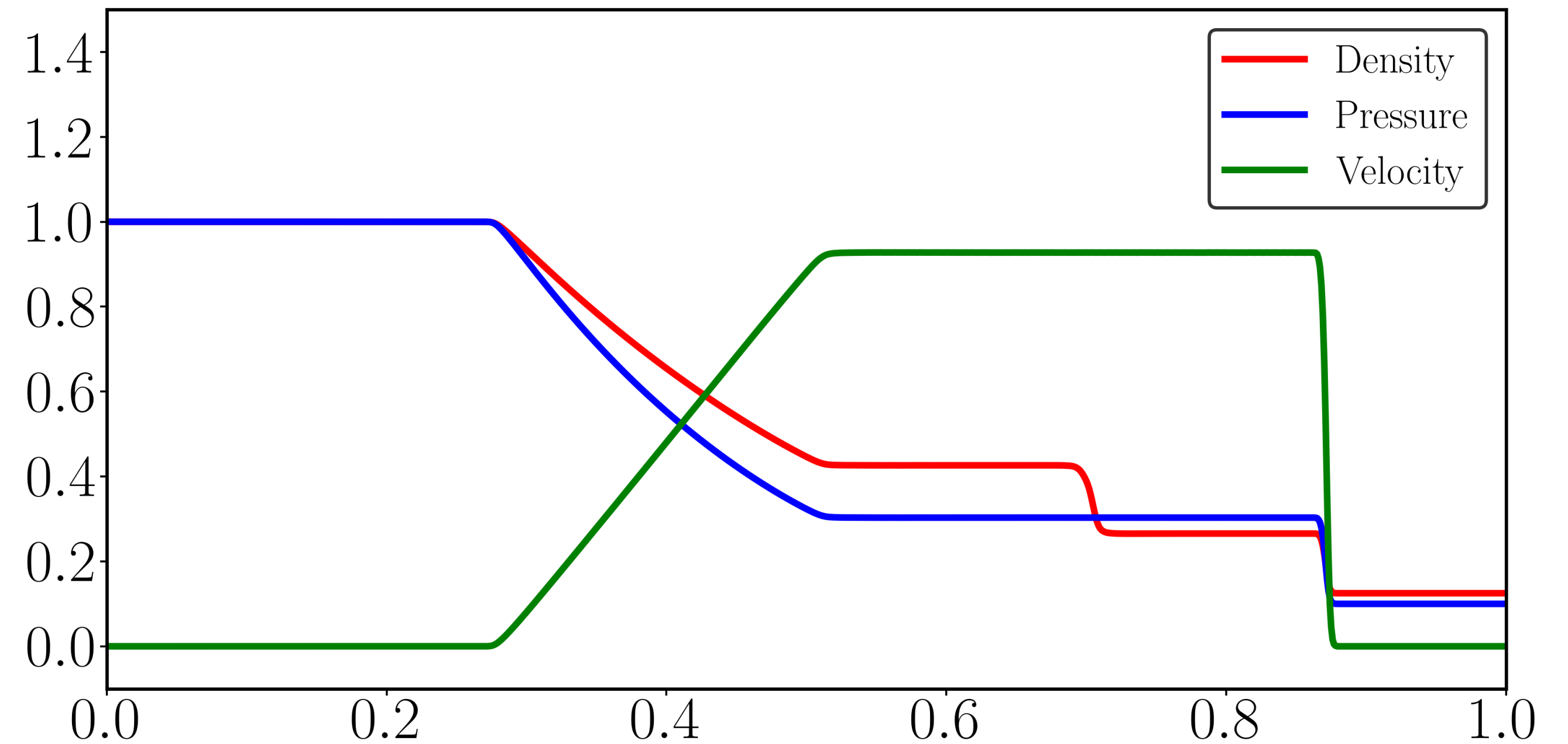
Second-order wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Simulation Sciences: PDEs

First-order hyperbolic equation: Euler equation for a compressible medium

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) &= 0 \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) &= -\frac{\partial p}{\partial x_i} \\ \frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_j}((\rho E + p)u_j) &= 0\end{aligned}$$

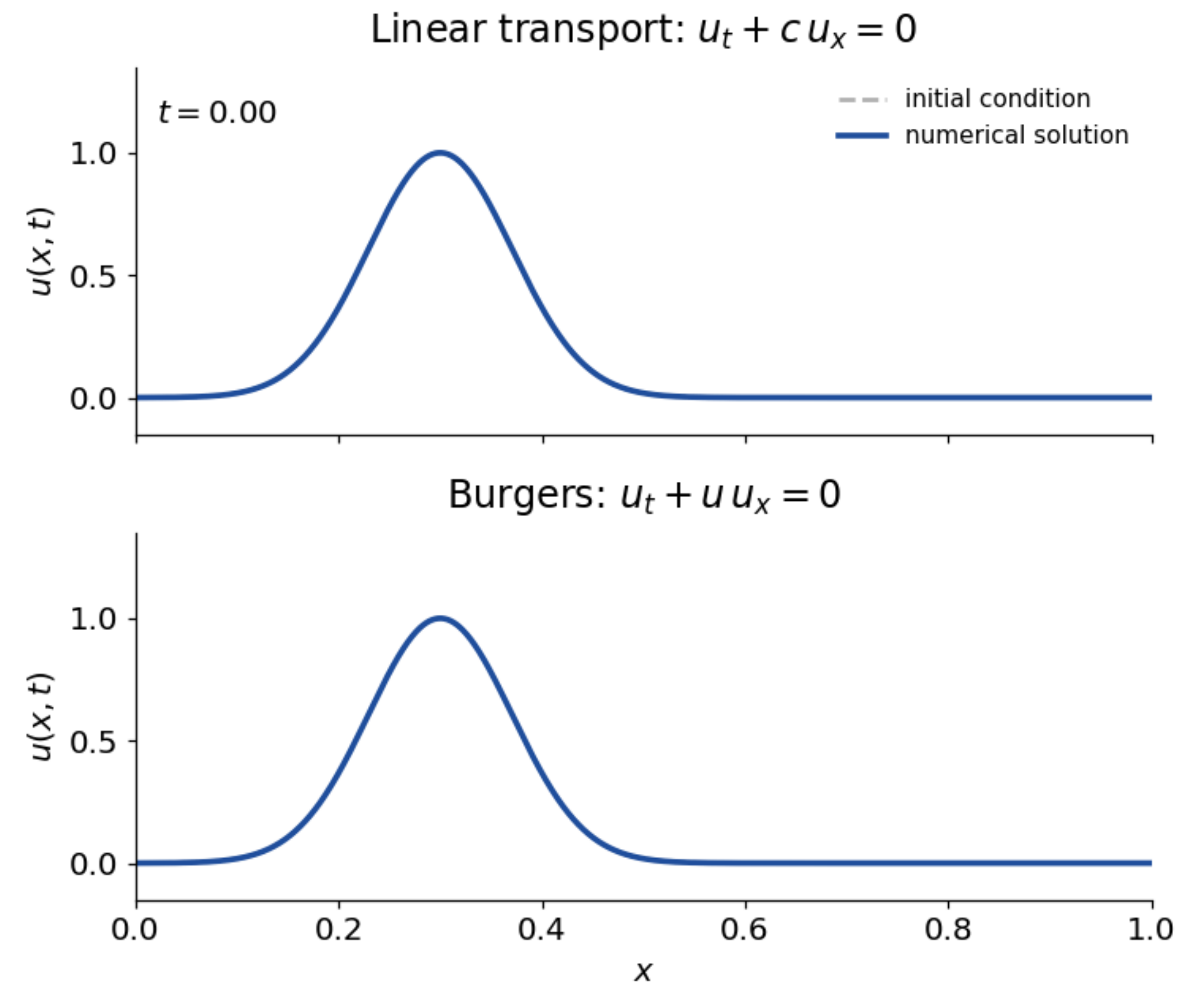


Simulation Sciences: PDEs (linear non-linear)

An equation is **linear** if the unknown and its derivatives appear only to the first power and are never multiplied together or passed through a nonlinear function, so that a sum of solutions is again a solution (superposition). In a **nonlinear** equation the solution feeds back on itself, for instance when the transport speed depends on the solution itself: new phenomena appear, such as shock formation.

Linear transport:
$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

Burgers:
$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$



Simulation Sciences: PDEs (strong vs weak form)

A **strong (classical) solution** is a function that is differentiable as many times as the equation requires and satisfies the PDE at every single point of the domain. This is a demanding requirement: it fails as soon as the solution has a kink or a jump, as happens with the shock in Burgers' equation or with a material interface in a heat conduction problem.

A **weak solution** relaxes the requirement: instead of asking the equation to hold pointwise, we multiply it by an arbitrary test function and integrate, so that it holds "on average" against all possible test functions.

Weak formulations matter for two reasons. They make it possible to describe physically meaningful solutions with discontinuities. And they are the foundation of the finite element method, where the weak form is projected onto a finite-dimensional space to obtain a linear system.

Strong form

$$-\frac{d^2 u}{dx^2} = f \quad \text{in } (0, 1), \quad u(0) = u(1) = 0$$

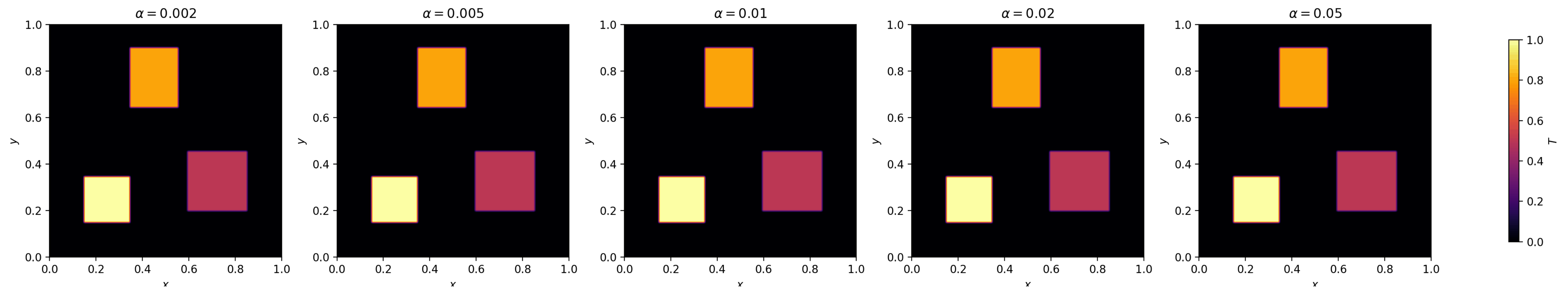
Weak form

$$\text{Find } u \in H_0^1(0, 1) \text{ such that } \int_0^1 \frac{du}{dx} \frac{dv}{dx} dx = \int_0^1 f v dx \quad \forall v \in H_0^1(0, 1)$$

Simulation Sciences: PDEs

In simplified settings (linear equations, constant coefficients, simple geometries), PDEs can be solved analytically, for example with **separation of variables** or the **method of characteristics**. Even when a full solution is out of reach, analytical reasoning gives useful insight into the overall behaviour, such as propagation speeds, decay rates or characteristic scales.

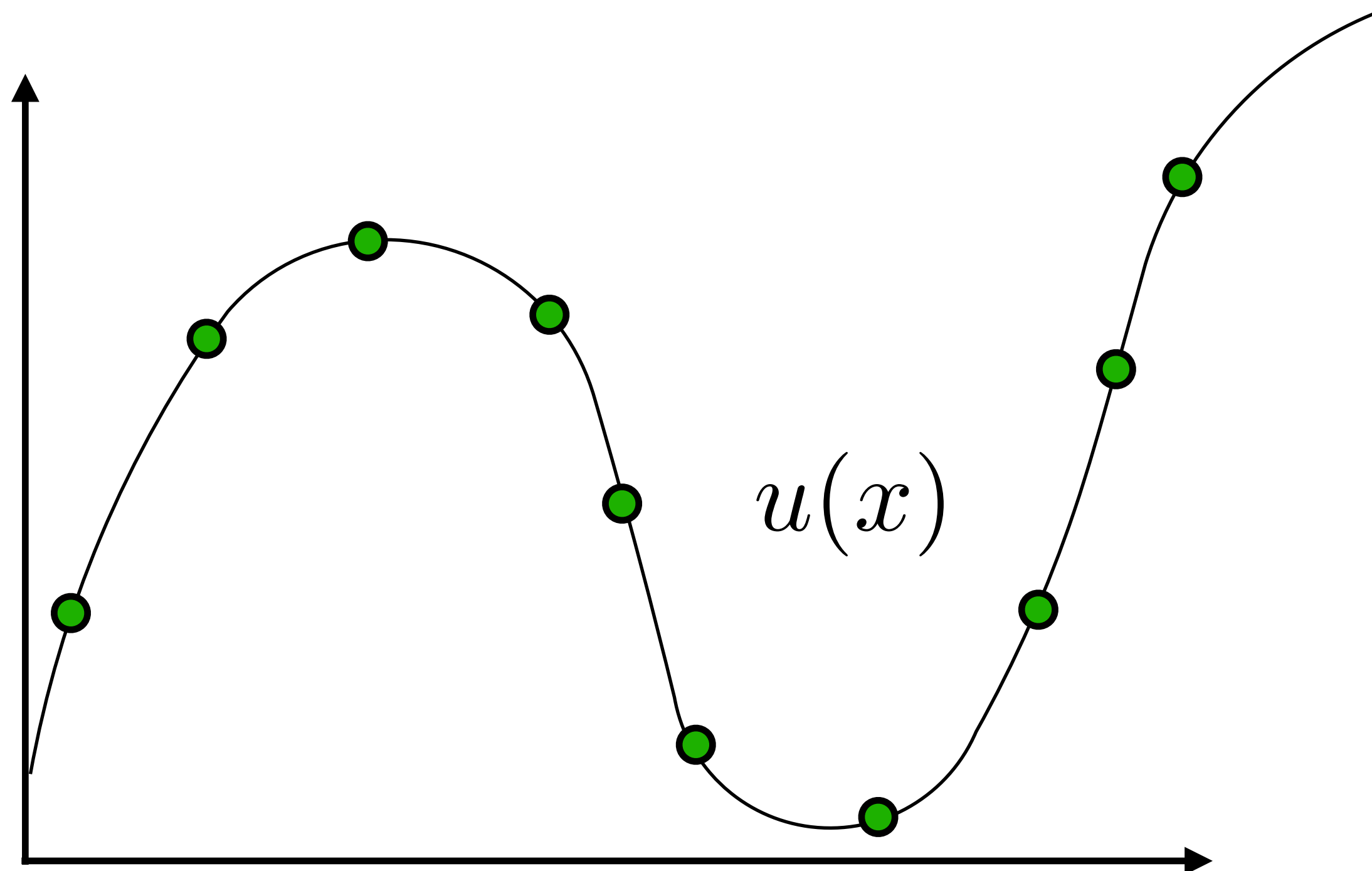
Realistic applications rarely fit this picture. Nonlinearity, complex geometries, heterogeneous materials and multiscale physics rule out closed-form solutions, so the PDE is replaced by a **discretised problem** that approximates the true one (**finite differences, finite volumes, finite elements**). These methods are very general, but their cost grows quickly with the resolution required, which motivates reduced-order models, PINNs and neural operators.



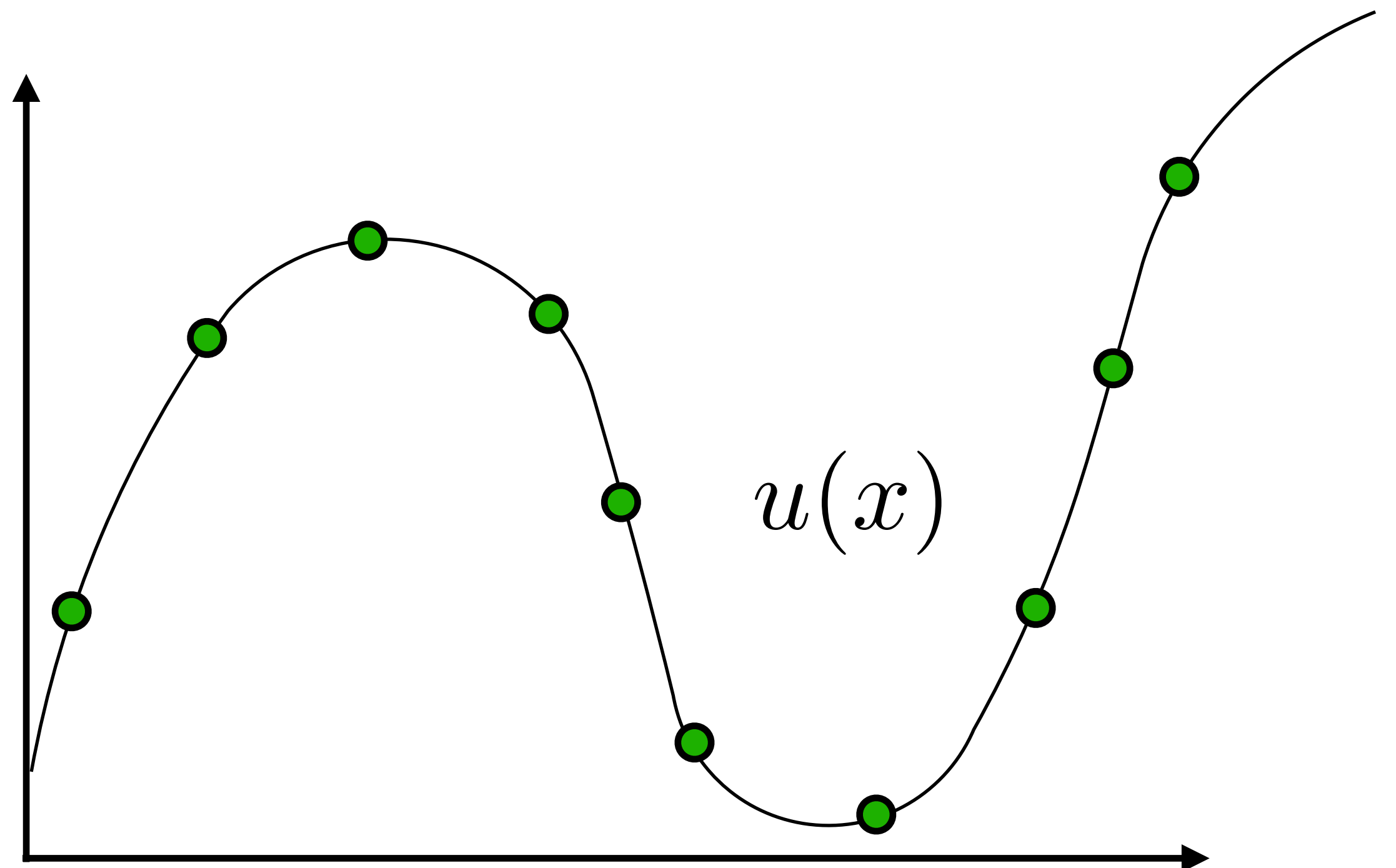
Simulation Sciences: PDEs

In general, PDEs live in infinite-dimensional function spaces: a solution is a function with suitable regularity properties, such as having a certain number of derivatives well defined pointwise (the space of k times continuously differentiable functions, $C^k(\Omega)$), or well defined in an integral sense (Sobolev spaces such as $H^k(\Omega)$), whose functions and weak derivatives up to order k are square-integrable). The first is the natural setting for strong solutions, the second for weak solutions.

Discretize



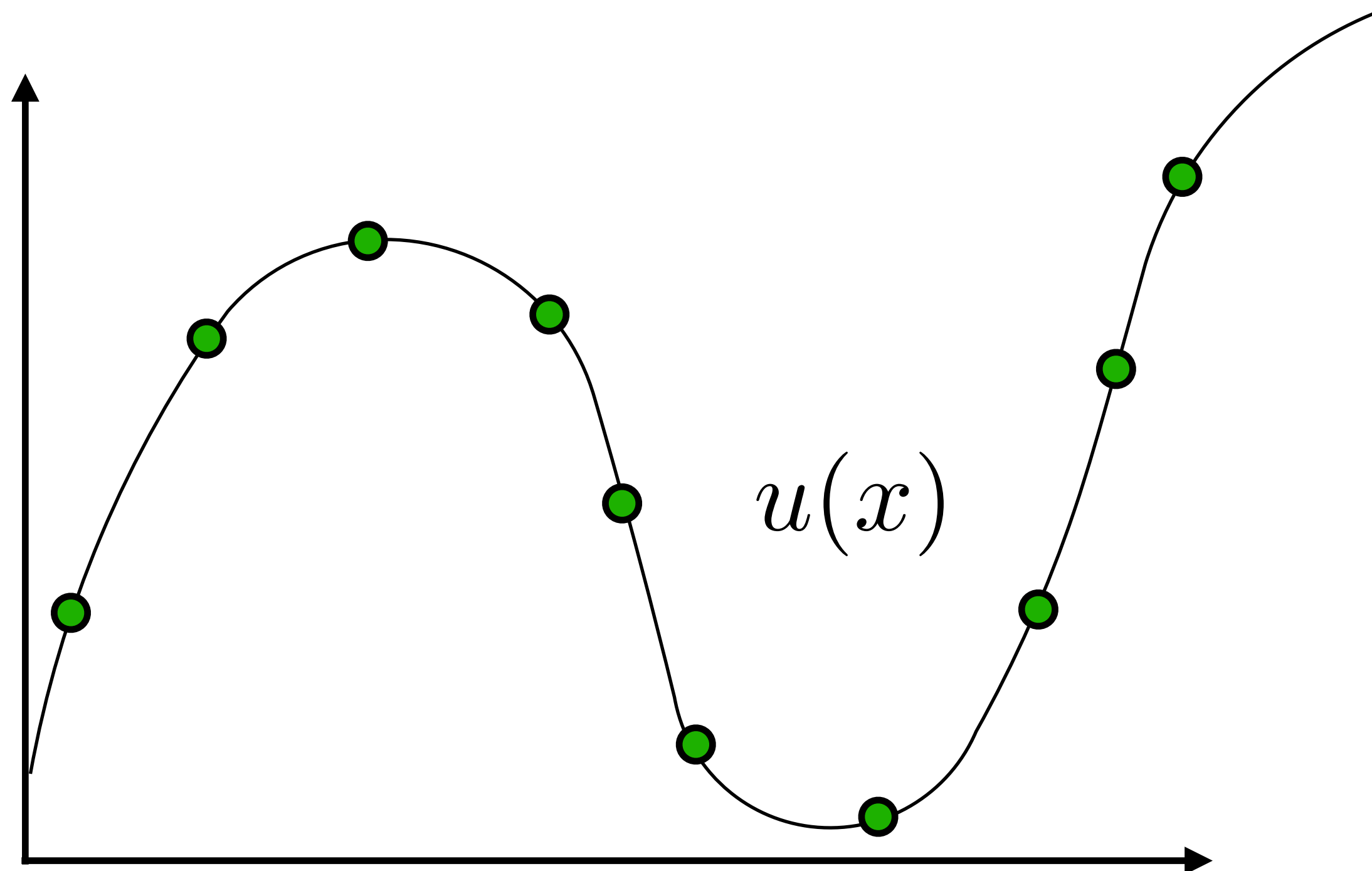
Compute derivatives



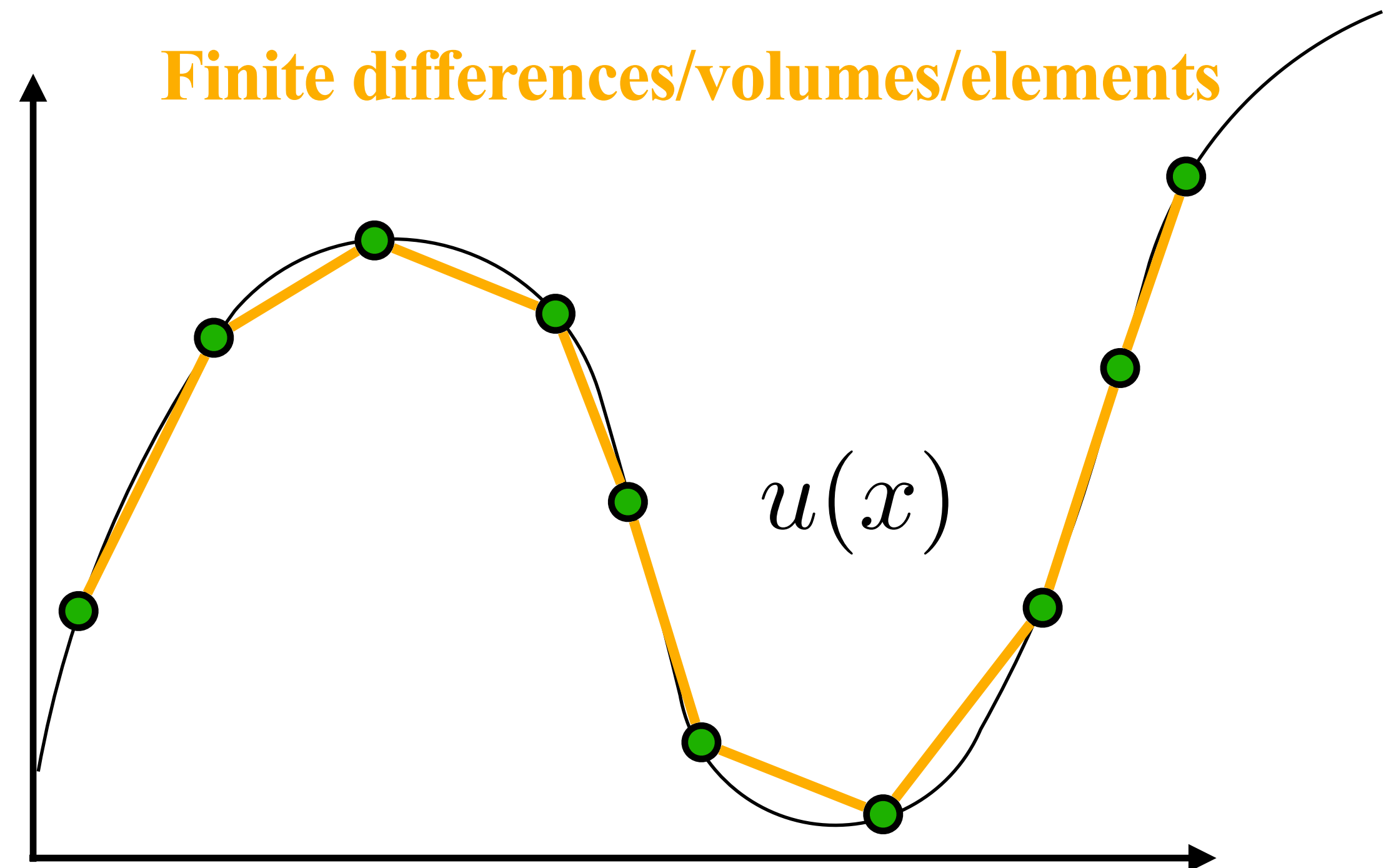
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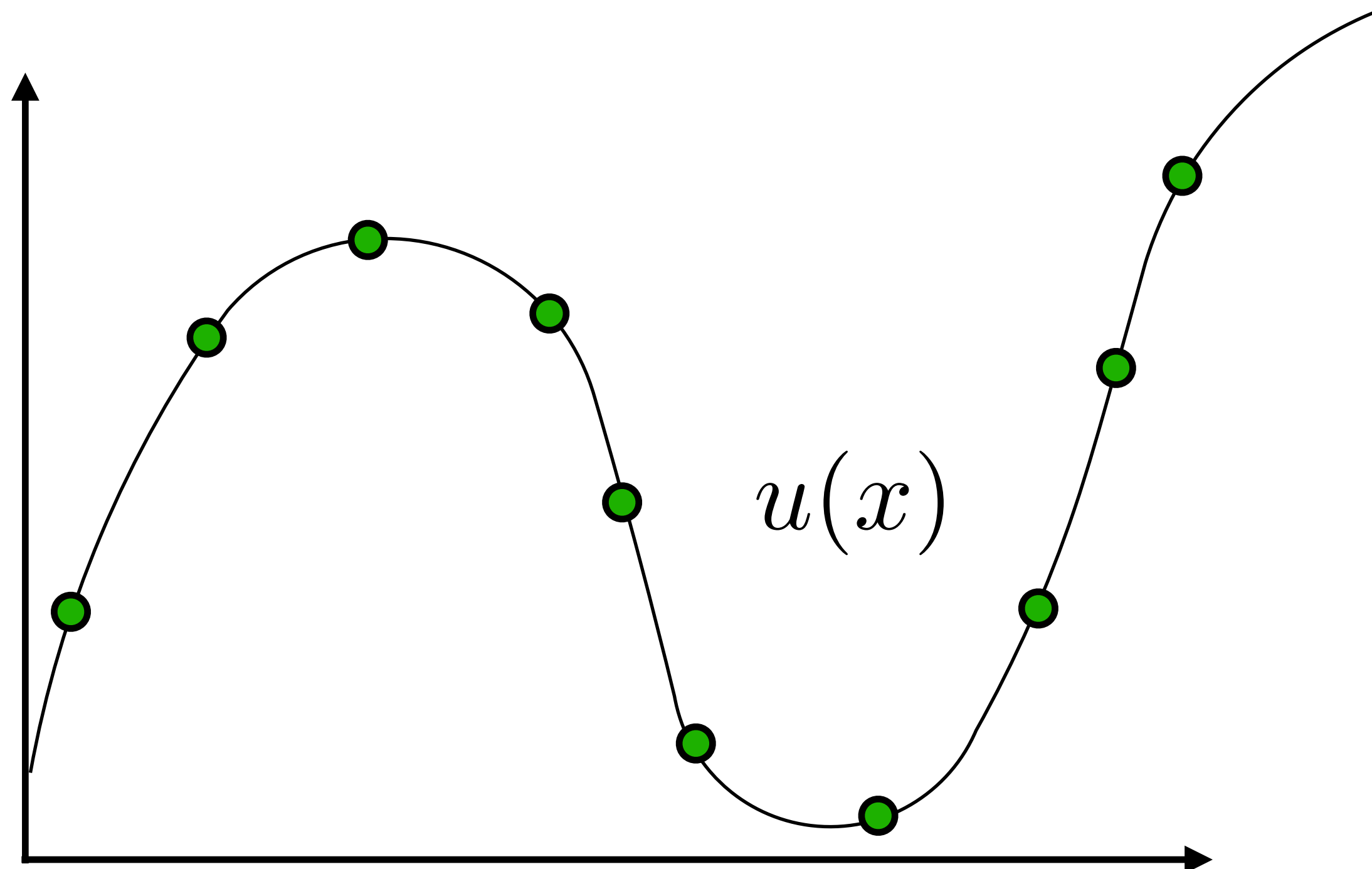
Compute derivatives



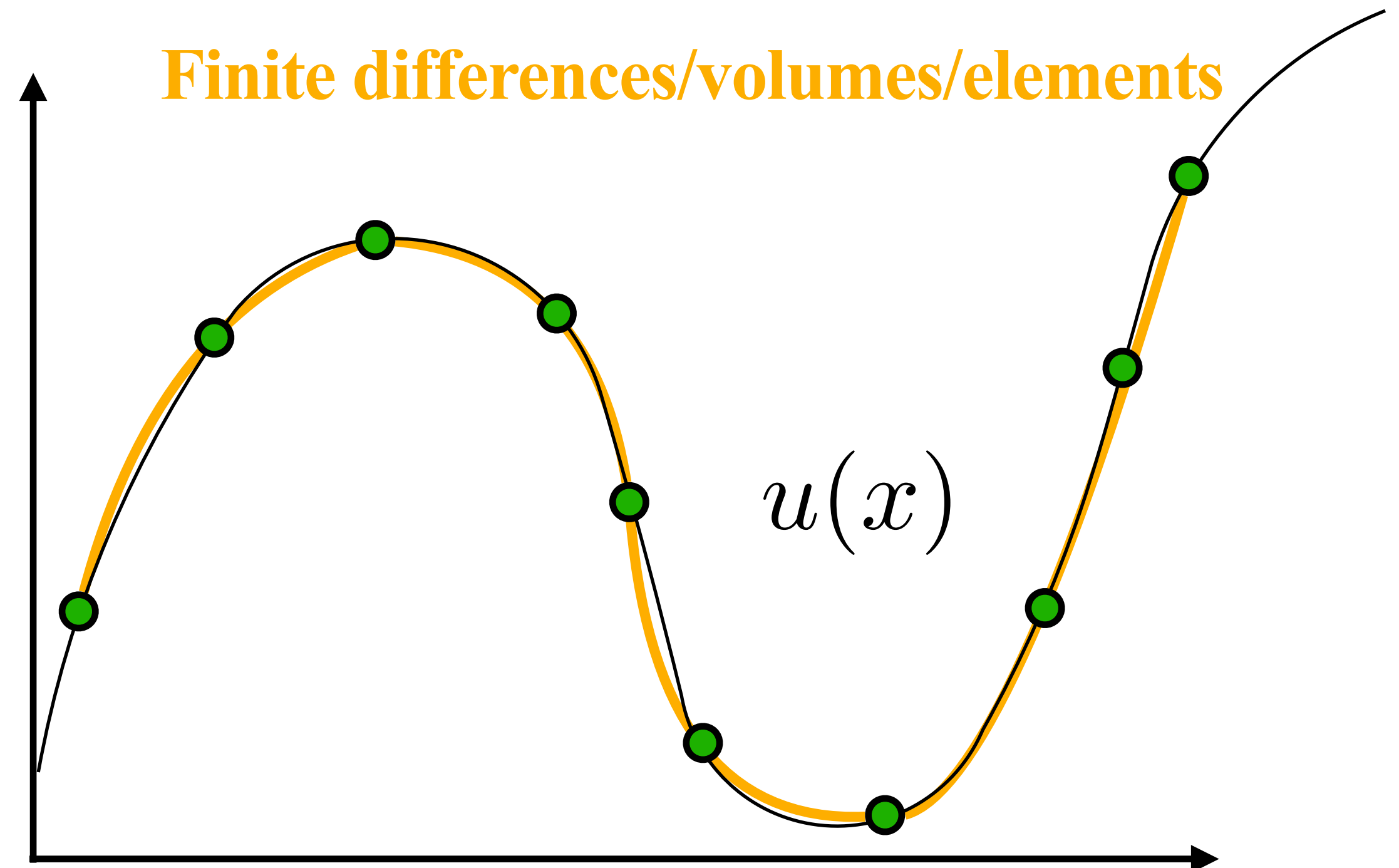
Simulation Sciences: PDEs

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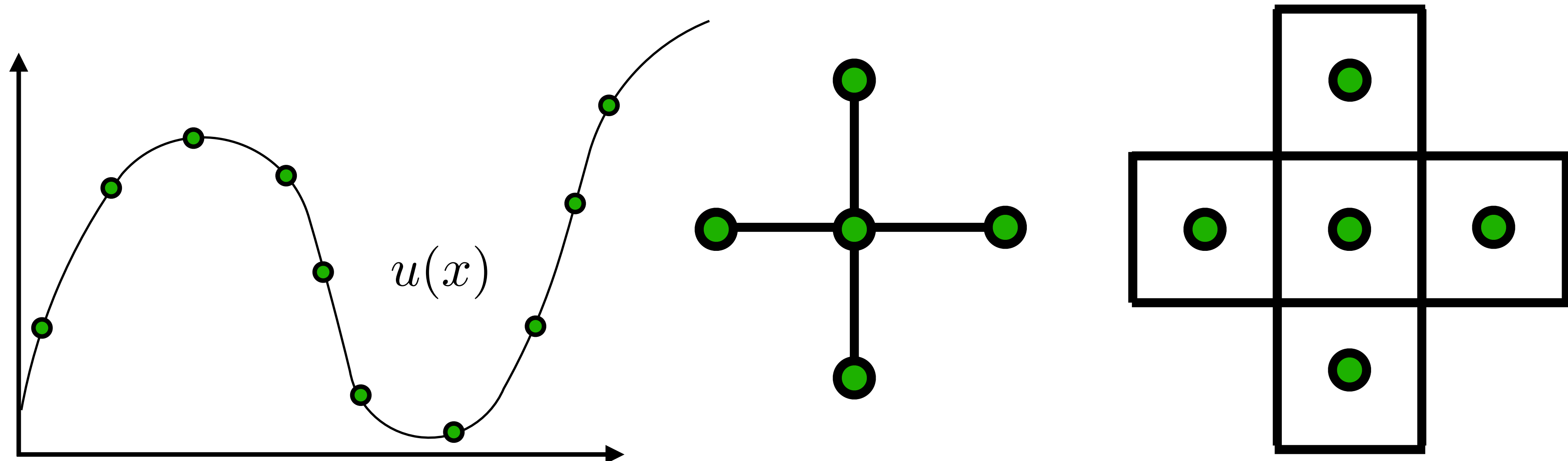
Compute derivatives



Simulation Sciences: PDEs

In one dimension, finite differences, finite volumes and finite elements are almost indistinguishable: on a uniform grid they often lead to the same stencil and the same tridiagonal system. The differences emerge in higher dimensions and on complex meshes. **Finite differences** need structured grids and struggle with irregular boundaries. **Finite volumes**, based on local conservation over cells, handle unstructured meshes naturally and are the standard choice in fluid dynamics. **Finite elements**, based on the weak form and local polynomial bases, offer the greatest geometric flexibility, at the price of heavier meshing and assembly.

In higher dimensions, mesh generation also becomes a major part of the workflow, and the number of degrees of freedom grows, which motivates reduced-order and mesh-free learned models.



Simulation Sciences: PDEs

FEM

Let's then consider our heat equation in the simplest configuration (1D) and we want to discretise this using the Finite Element Method:

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

With FEM we start from the weak-formulation: we test the PDE against a test function and we then integrate:

$$\int_0^1 v \frac{\partial u}{\partial t} dx - \kappa \int_0^1 v \frac{\partial^2 u}{\partial x^2} dx = 0.$$

Now, we are still living on the continuum level: we have not introduced any discretisation and (our solution is infinite dimensional). So, we look for a subset of our functional space: we look for solutions of the type

$$u(x) = \sum_{i=1}^N u_i \phi_i(x)$$

Simulation Sciences: PDEs

FEM

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Simulation Sciences: PDEs

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$$u(x) = \sum_{i=1}^N u_i \phi_i(x)$$

For a general set of functions that we call basis functions. We then also set the test functions of the weak formulation to be like these ones and we get:

$$\int_0^1 \phi_j \frac{\partial}{\partial t} \left(\sum_{i=1}^N u_i \phi_i(x) \right) dx - \kappa \int_0^1 \phi_j \frac{\partial^2}{\partial x^2} \left(\sum_{i=1}^N u_i \phi_i(x) \right) dx = 0, \quad \forall j = 1, \dots, N$$

And by simplifying and integrating by parts we get:

$$\sum_{i=1}^N \frac{du_i}{dt} \int_0^1 \phi_j \phi_i dx - \kappa \sum_{i=1}^N u_i \left[\phi_j \frac{d\phi_i}{dx} \right]_0^1 + \kappa \sum_{i=1}^N u_i \int_0^1 \frac{d\phi_j}{dx} \frac{d\phi_i}{dx} dx = 0. \quad \forall j = 1, \dots, N$$

Simulation Sciences: PDEs

FEM

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Simulation Sciences: PDEs

FEM

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Now, since we are lazy we simply consider the Neumann homogeneous boundary conditions and we get rid of the boundary terms:

$$\sum_{i=1}^N \frac{du_i}{dt} \boxed{\int_0^1 \phi_j \phi_i dx} + \kappa \sum_{i=1}^N u_i \boxed{\int_0^1 \frac{d\phi_j}{dx} \frac{d\phi_i}{dx} dx} = 0. \quad \forall j = 1, \dots, N$$

M_{ij} K_{ij}

We get the system of ODEs:

$$\mathbf{M} \frac{d\mathbf{u}}{dt} = -\kappa \mathbf{K} \mathbf{u}$$

Simulation Sciences: PDEs

FEM

A few relevant comments for this course:

The mapping between the parameter κ and the solution u passes through the resolution of a (possibly) very high-dimensional system of ODEs

A possible objective of SciML might be getting the mapping

$$\kappa \rightarrow u$$

in a much much faster way (this can be achieved with **Reduced-Order Models** or **Neural Operators**)

Simulation Sciences: PDEs

FEM

A few relevant comments for this course:

When we pass from an infinite dimensional space to a discrete one we use a **basis**:

$$u(x) = \sum_{i=1}^N u_i \phi_i(x)$$

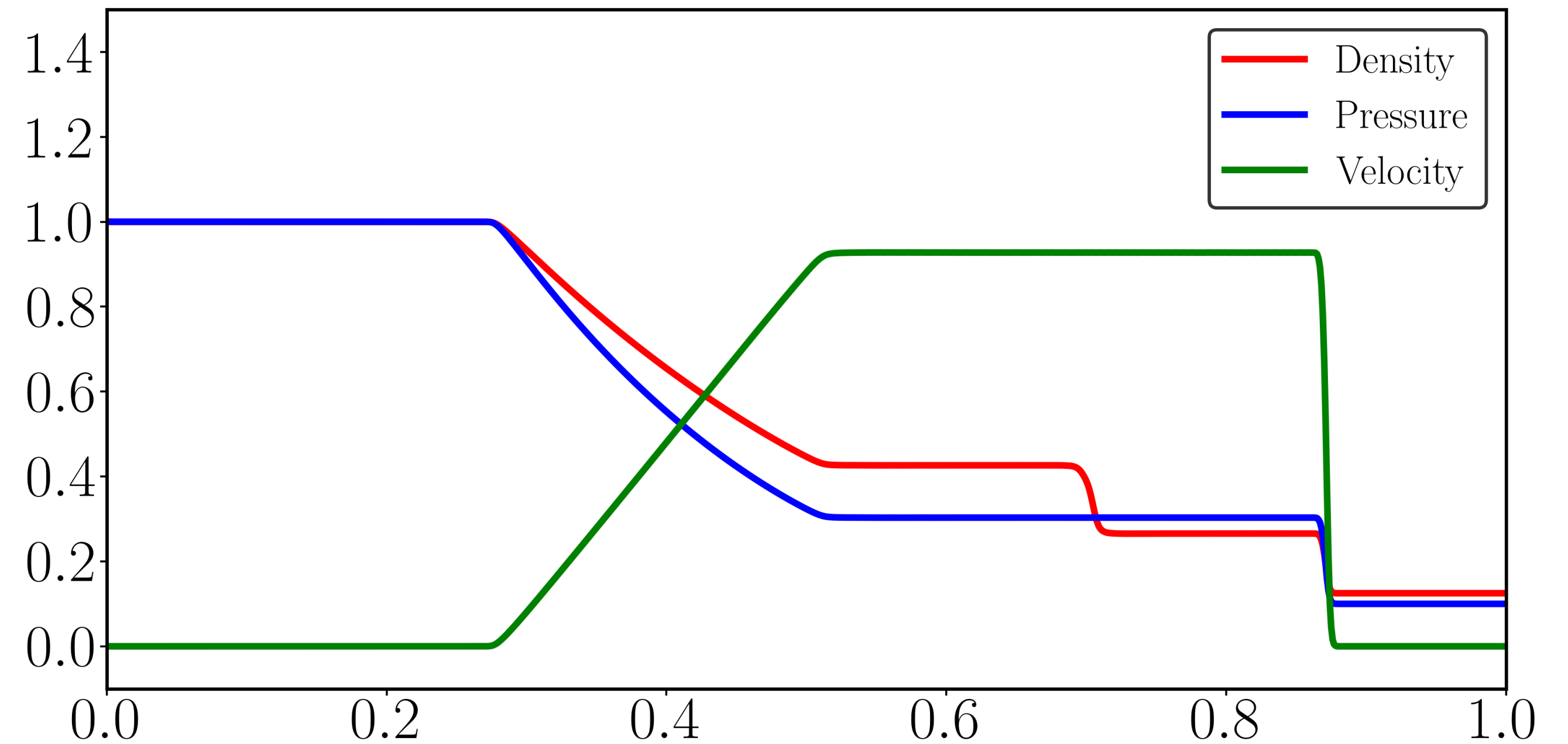
We never really specified this but common choice might be polynomials or polynomials restricted to certain regions, but, they could also be **Neural Networks** (this connects with the so-called **Physics-Informed Neural Networks**)

Simulation Sciences: challenges

Here I showed some very simple PDEs but of course the complexity can increase significantly for problems of engineering interest:

Euler equations

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) &= 0 \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) &= -\frac{\partial p}{\partial x_i} \\ \frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_j}((\rho E + p)u_j) &= 0\end{aligned}$$

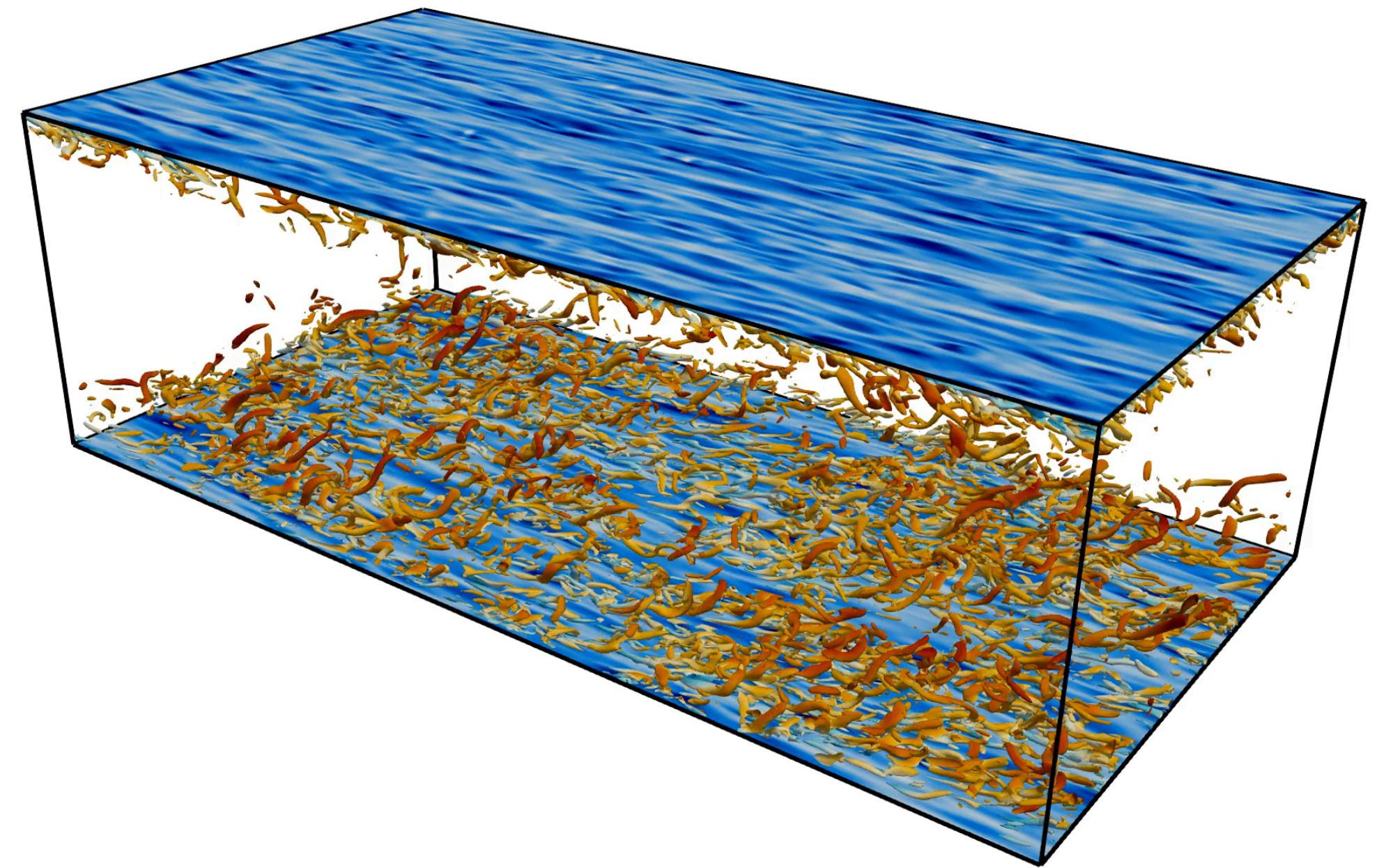


Simulation Sciences: challenges

Here I showed some very simple PDEs but of course the complexity can increase significantly for problems of engineering interest:

Incompressible Navier-Stokes equations

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = - \frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j}$$
$$\frac{\partial u_j}{\partial x_j} = 0$$



Simulation Sciences: challenges

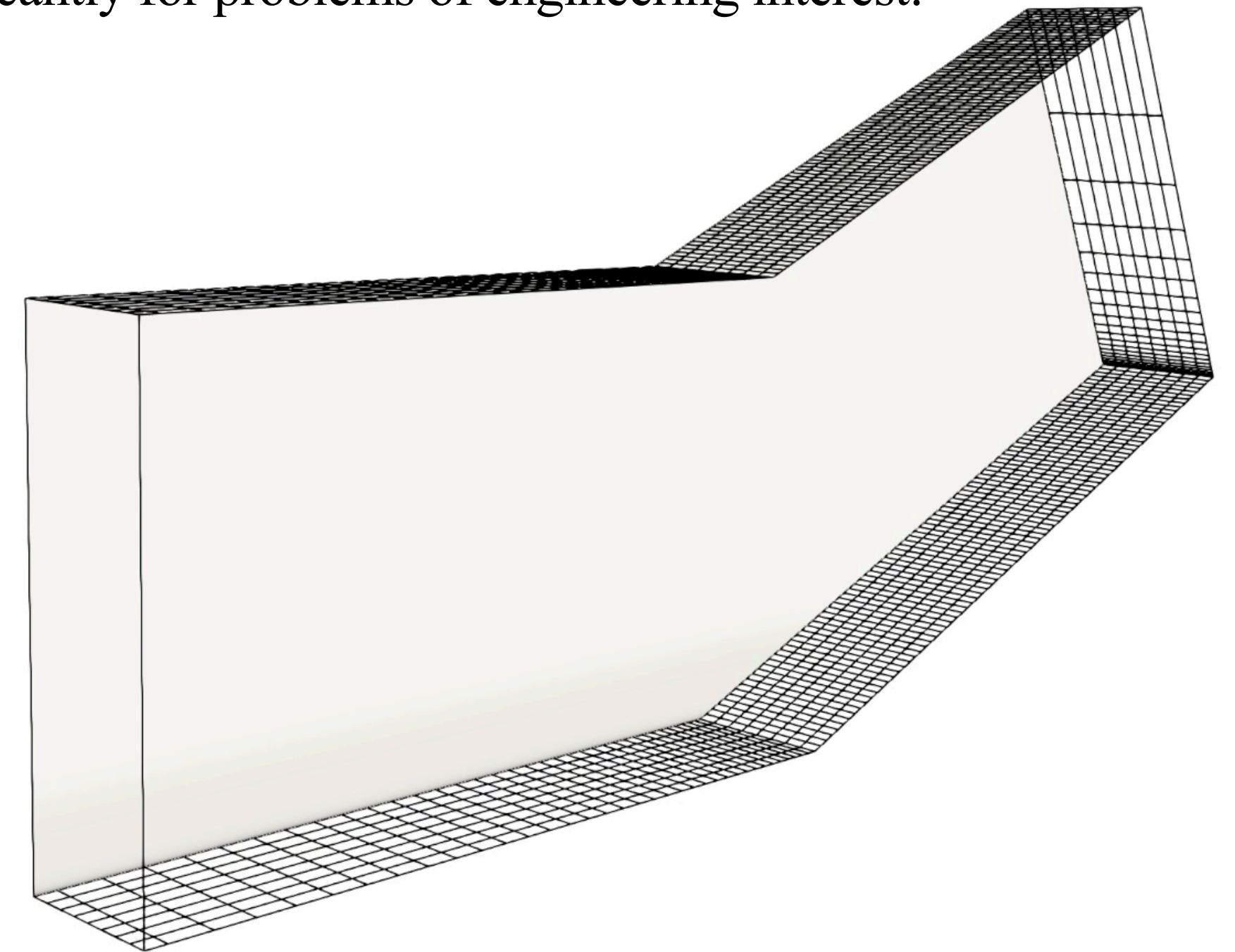
Here I showed some very simple PDEs but of course the complexity can increase significantly for problems of engineering interest:

Compressible Navier-Stokes equations

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) = 0$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial S_{ij}}{\partial x_j} + \rho g_i$$

$$\frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_j}((\rho E + p)u_j) = \frac{\partial S_{ij} u_i}{\partial x_j} + \rho g_i u_i$$

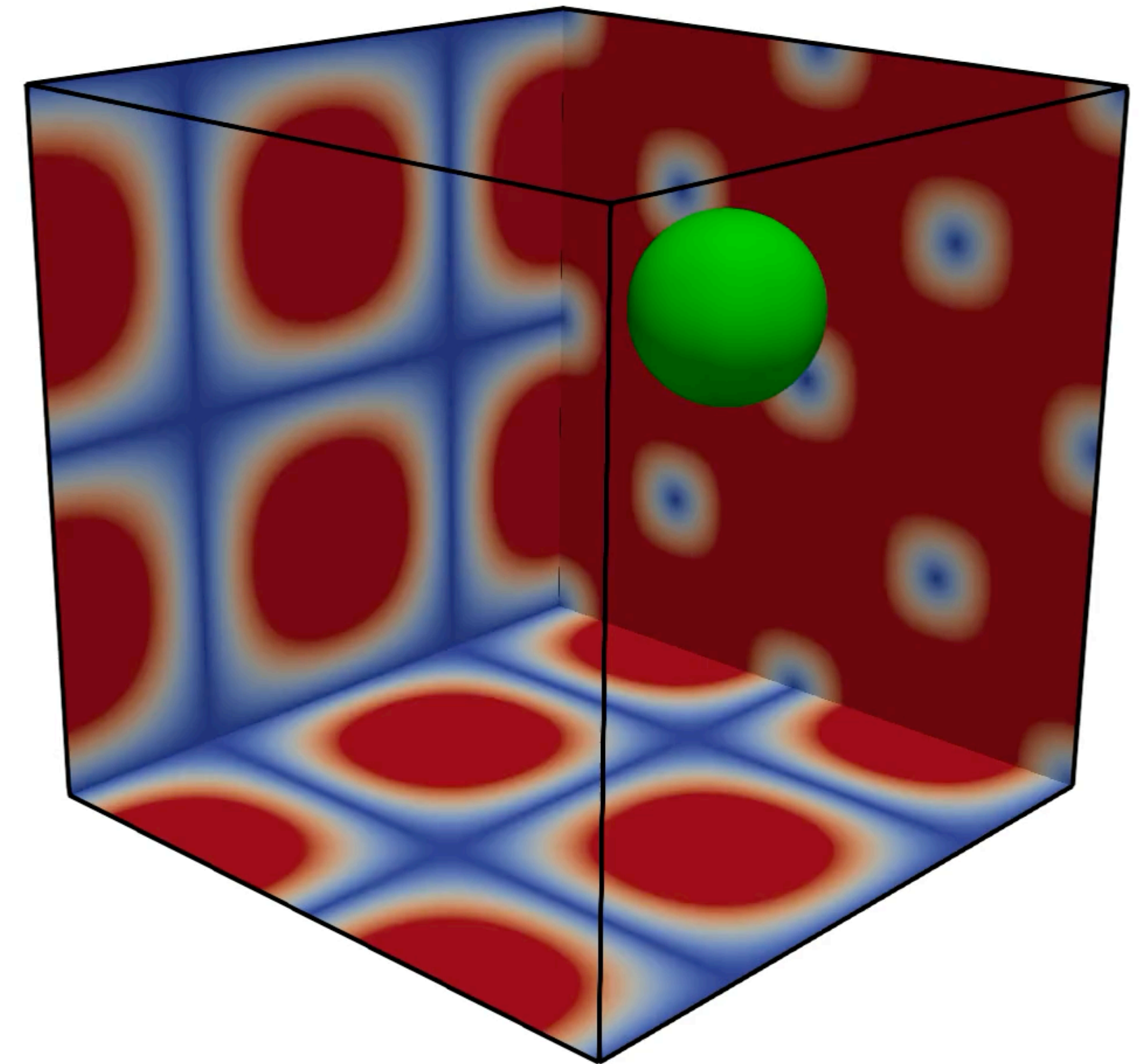


Simulation Sciences: challenges

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$$\begin{aligned}\frac{\partial \phi}{\partial t} + u_j \frac{\partial \phi}{\partial x_j} &= 0 \\ \frac{\partial \rho_2}{\partial t} + \frac{\partial}{\partial x_j} (\rho_2 u_j) &= 0 \\ \frac{\partial \rho_1}{\partial t} + \frac{\partial}{\partial x_j} (\rho_1 u_j) &= 0 \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_i u_j) &= -\frac{\partial p}{\partial x_i} + \frac{\partial S_{ij}}{\partial x_j} + \rho g_i \\ \frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_j} ((\rho E + p) u_j) &= \frac{\partial S_{ij} u_i}{\partial x_j} + \rho g_i u_i\end{aligned}$$

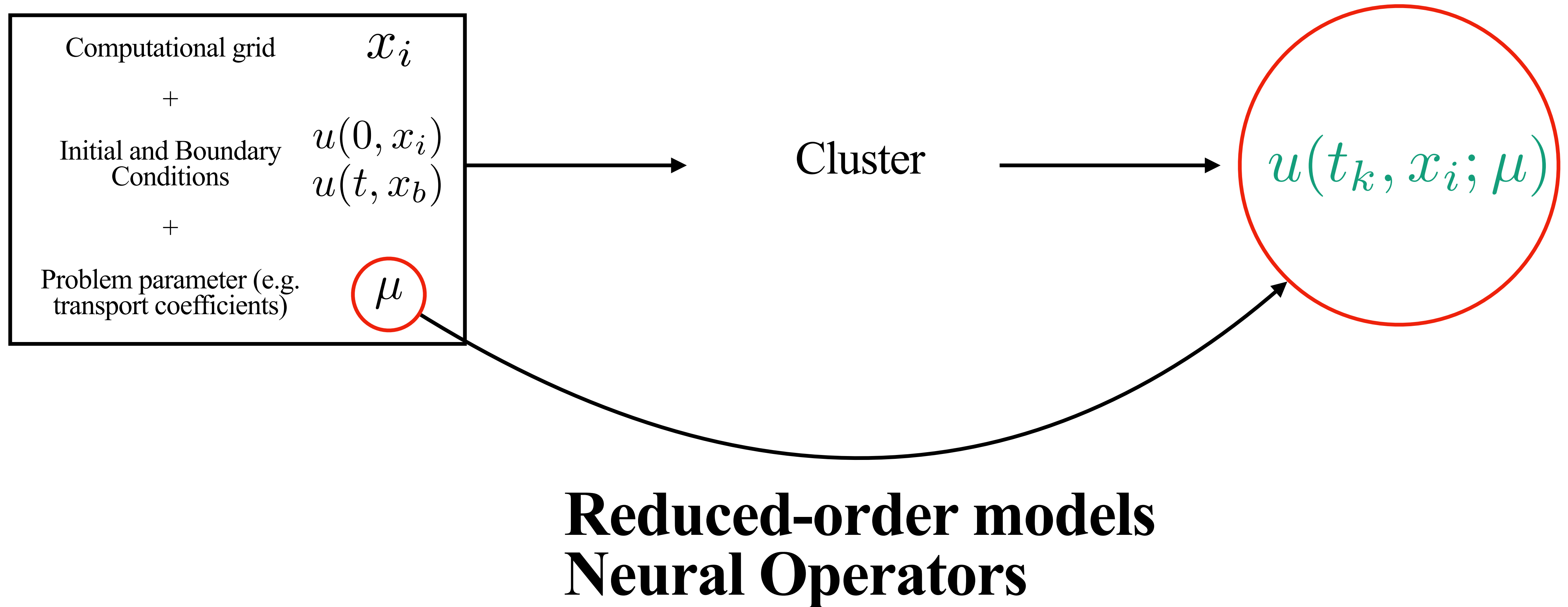
**Baer-Nunziato
five-equation
model**



Simulation Sciences: the role of ML

What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

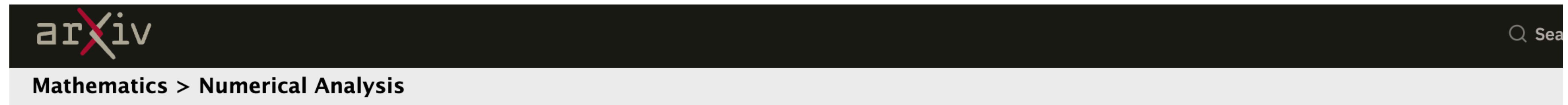
Speed-up simulations:



Simulation Sciences: the role of ML

What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

Speed-up simulations:



[Submitted on 16 Mar 2026]

Efficient and Accurate Surrogate Modeling of Turbulent Flows via Space-Dependent Aggregation and Reduced Order Models

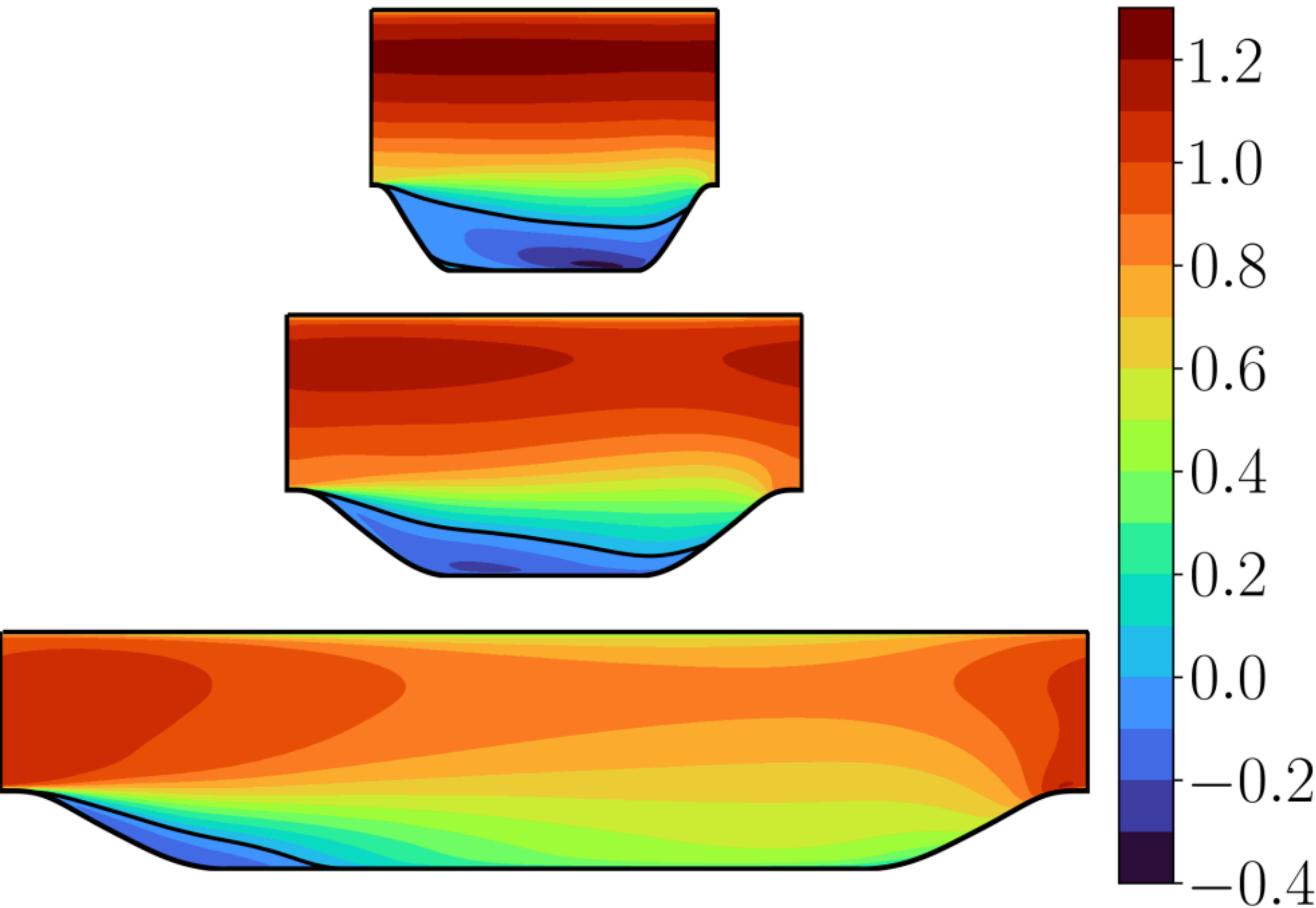
Piero Zappi, Anna Ivagnes, Niccolò Tonicello, Gianluigi Rozza

Reynolds-Averaged Navier-Stokes (RANS) models are widely used for turbulent flow simulations due to their computational efficiency, but their accuracy strongly depends on the selected turbulence closure and may vary across the flow domain. Space-dependent model aggregation has been shown to improve RANS predictions by combining multiple turbulence models, although at the cost of repeated high-fidelity simulations. The first novelty of this work is a unified framework that combines different turbulence models, space-dependent aggregation, and non-intrusive reduced order models to achieve both accuracy and efficiency. Two aggregation pipelines are proposed: a Mixed FOM-ROM (MFR) approach, where a reduced order model is trained on aggregated RANS solutions, and a Mixed-ROM (MR) approach, which directly aggregates multiple reduced order models built on top of different RANS full-order models. The second novelty is that the aggregation weights are learned via a neural-network that provides smooth, space-continuous weights and improves generalization with respect to standard weighting techniques. The resulting surrogate models are validated on the two-dimensional periodic hill benchmark and on the flow over a height-dependent bump, demonstrating improved accuracy over individual RANS and ROM predictions at near real-time computational cost.

Simulation Sciences: the role of ML

What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

Speed-up simulations:



Objective:

We have parametrised geometries and we cannot afford to run a new simulation every time we change the geometry. We need a **fast** solution for **new parameters**

arXiv

Mathematics > Numerical Analysis

Submitted on 16 Mar 2026]

Efficient and Accurate Surrogate Modeling of Turbulent Flows via Space-Dependent Aggregation and Reduced Order Models

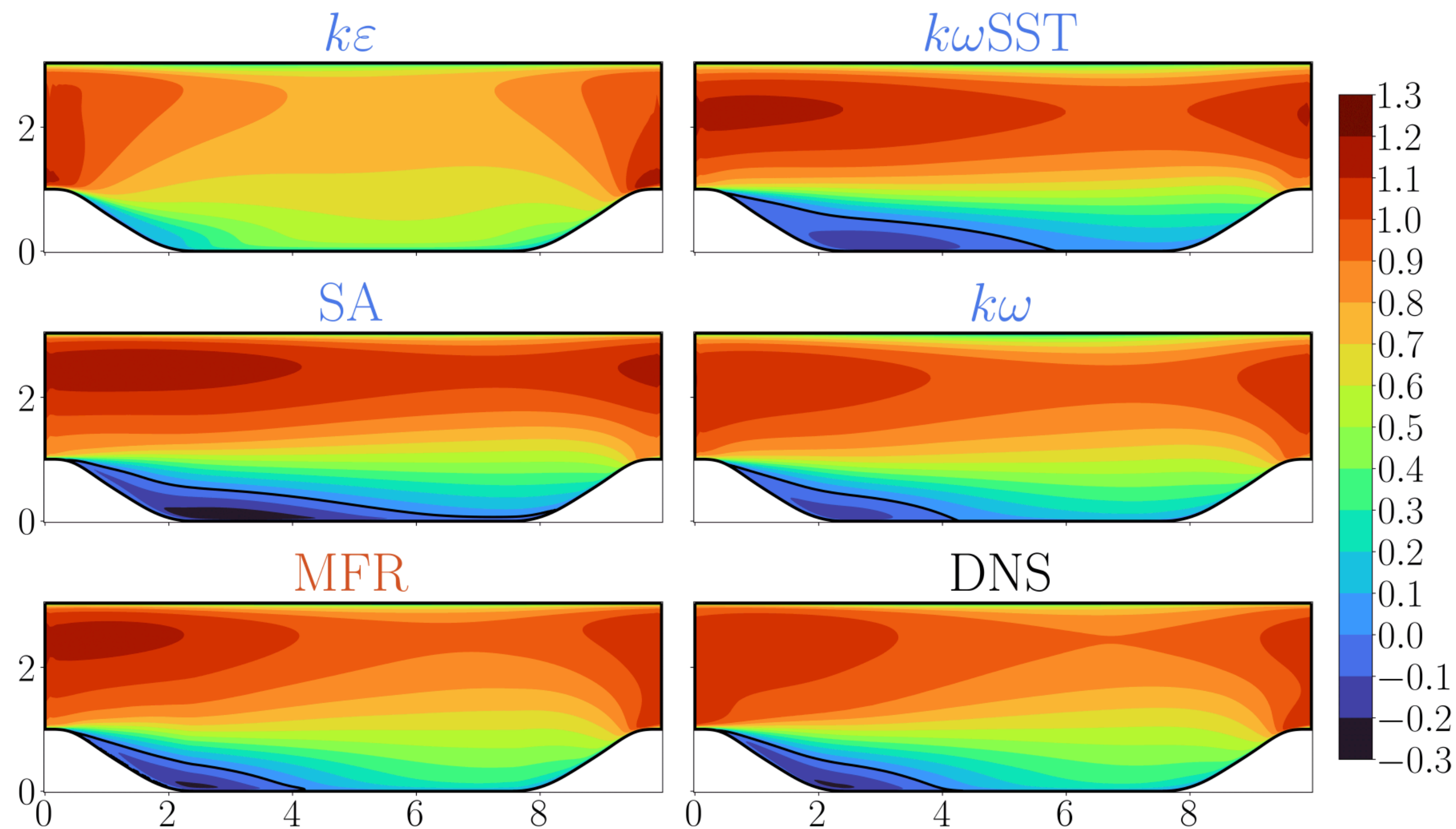
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Objective:

On top of that, different **RANS models** provide different result

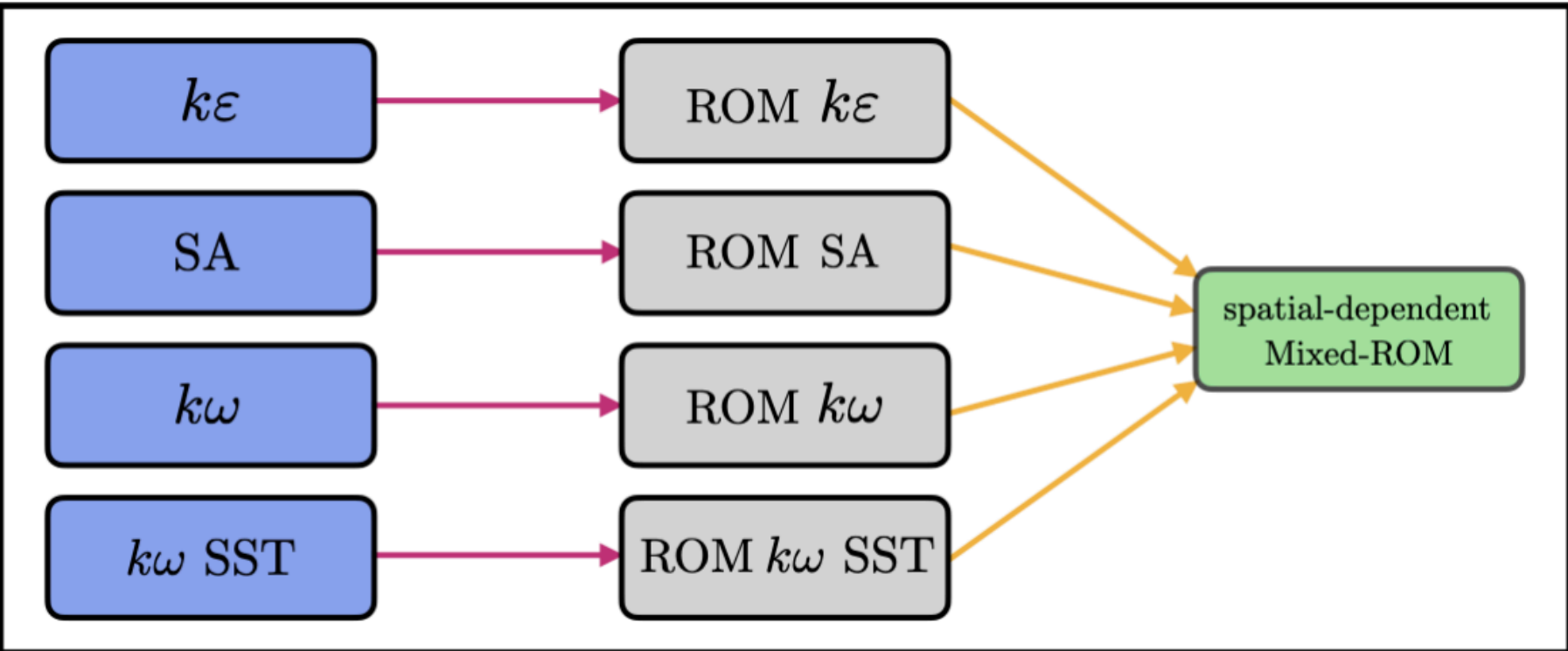
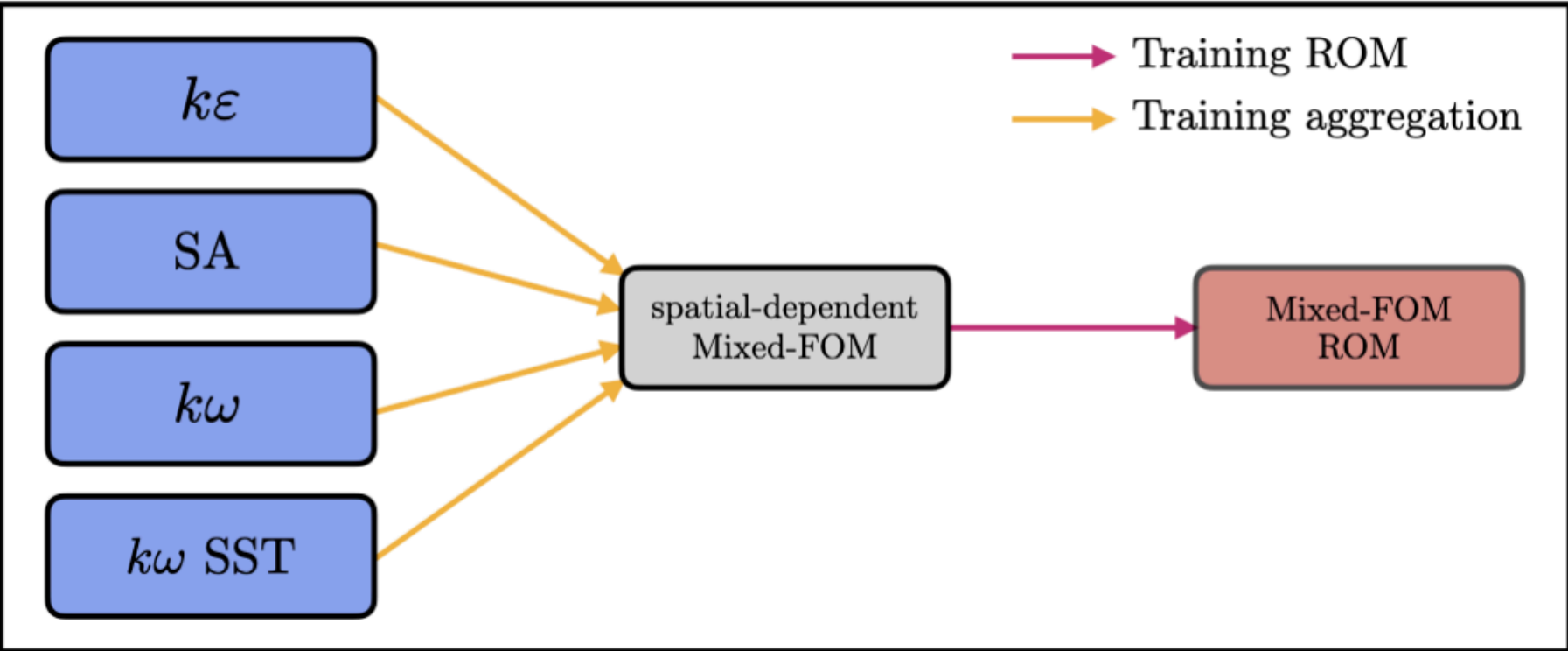
Question:

Can we optimally **spatially** combine the models in order to get a better overall solution?

Simulation Sciences: the role of ML

What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

Speed-up simulations:



Question:

can we do that without losing computational efficiency given by Reduced-Order Models?

arXiv

Mathematics > Numerical Analysis

Submitted on 16 Mar 2026]

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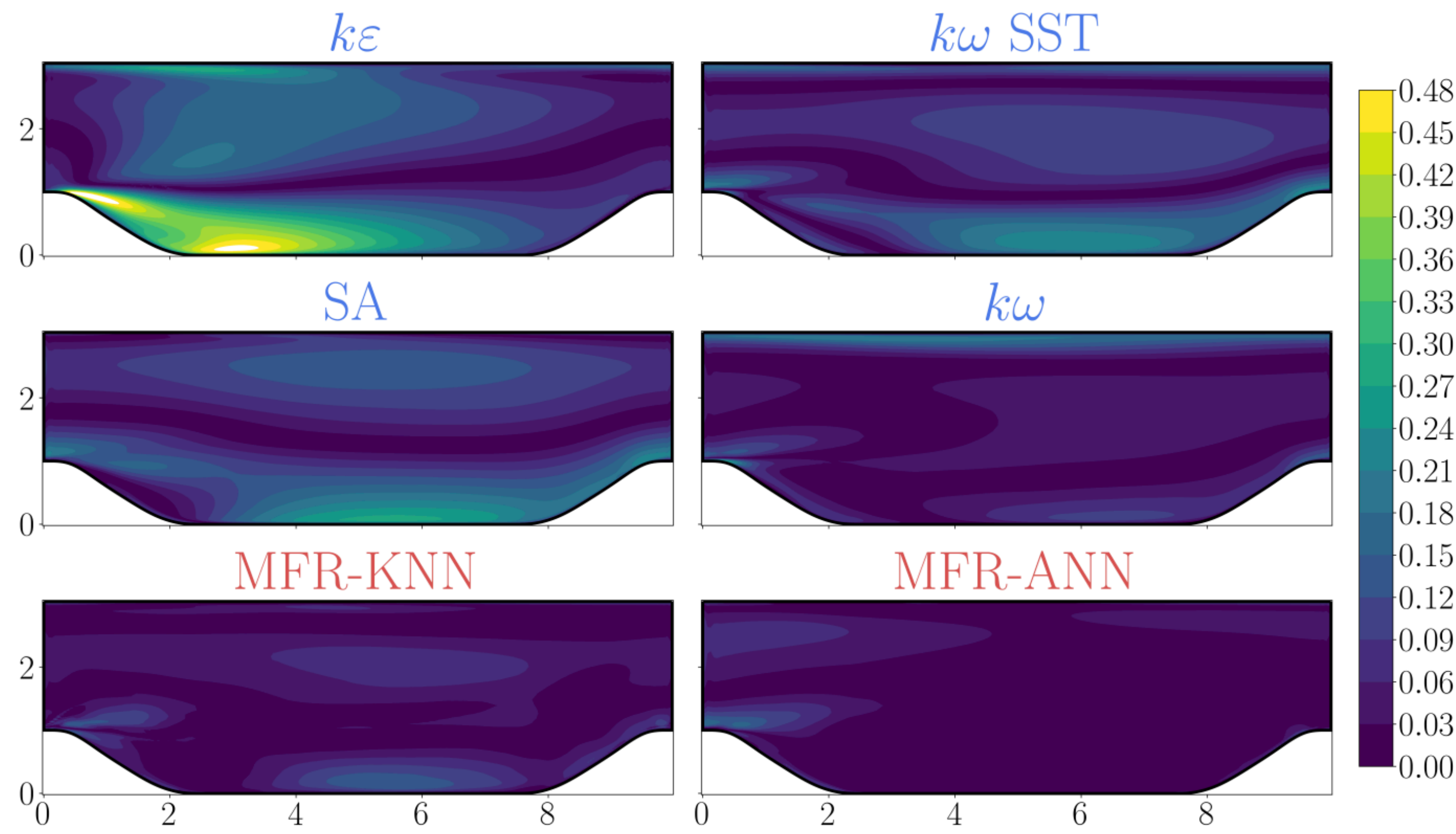
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Answer:

Indeed we can do that

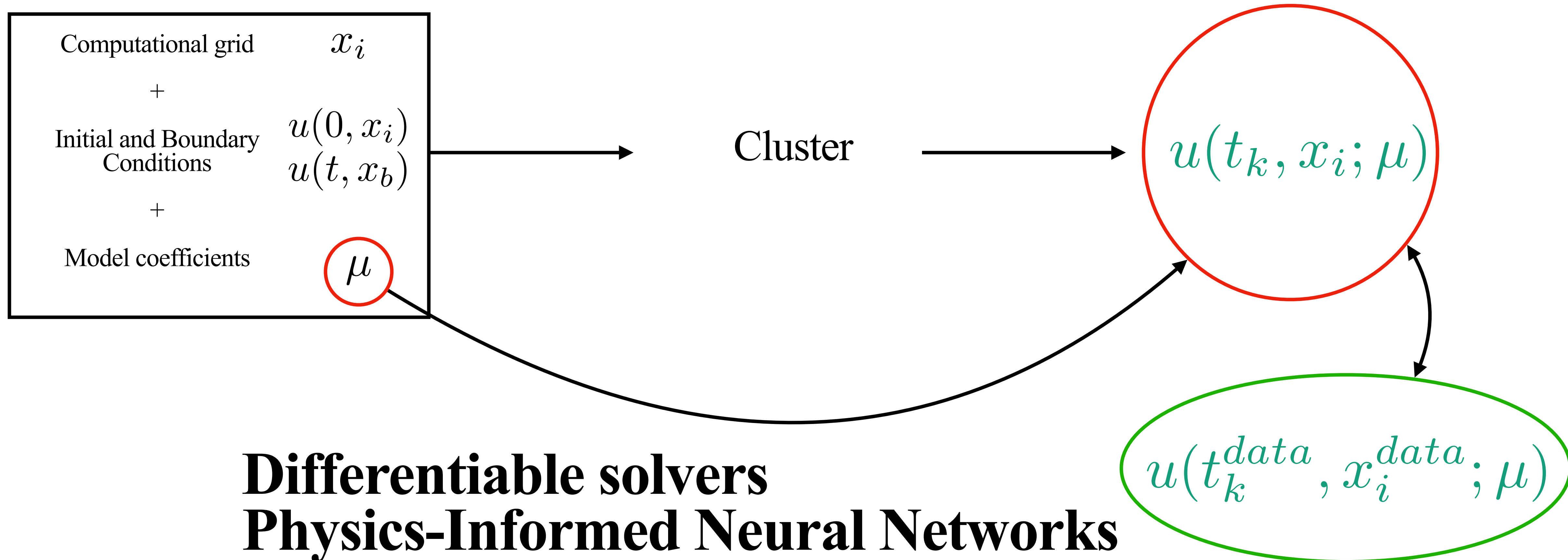
Drawbacks:

Same DoF for each simulation. These are just RANS but some of them cannot recover experiments anyway (multi fidelity?)

Simulation Sciences: the role of ML

What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

Calibrate models/inverse problems:



Simulation Sciences: the role of ML

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Physics > Fluid Dynamics

[Submitted on 31 May 2026]

End-to-end optimization of subgrid scale models for discontinuous spectral element schemes based on the discrete adjoint method

Nicola Clinco, Niccolò Tonicello, Paola Cinnella, Gianluigi Rozza

In computational fluid dynamics, Large Eddy Simulation (LES) offers a compelling balance between accuracy and computational cost by resolving large-scale flow structures while modeling unresolved subgrid scales. However, its predictive capacity is critically dependent on the choice and calibration of subgrid-scale (SGS) models, which often involve problem-dependent parameters and exhibit intricate interactions with the numerical discretization. In this work, we propose a discrete-adjoint framework to optimize SGS model parameters in the loop, leveraging automatic differentiation within a high-order Spectral Difference (SD) solver. Coarse-grained simulations of Forced Homogeneous Isotropic Turbulence (FHIT), together with filtered Direct Numerical Simulation (DNS) data, are used to optimize a limited set of parameters for classical SGS models, including the Smagorinsky model and non-linear tensor-basis formulations. For chaotic systems such as LES, the choice of objective function plays a crucial role in the stability and accuracy of the optimization. Here, we consider the spatio-temporally averaged decay of the Legendre modal coefficients as the quantity of interest for the SD scheme. The optimization is performed across different grid resolutions and polynomial orders, highlighting the impact of numerical discretization on model performance. The methodology is applied to both one-dimensional Burgers turbulence and fully three-dimensional turbulence. The trained models are subsequently assessed on out-of-sample configurations, including Decaying Homogeneous Isotropic Turbulence (DHIT) and the Taylor-Green vortex. Variations in polynomial order, grid resolution, and Reynolds number are considered to evaluate robustness and generalization. In all test cases, the optimized models demonstrate significant improvements over baseline SGS closures.

Simulation Sciences: the role of ML

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arXiv

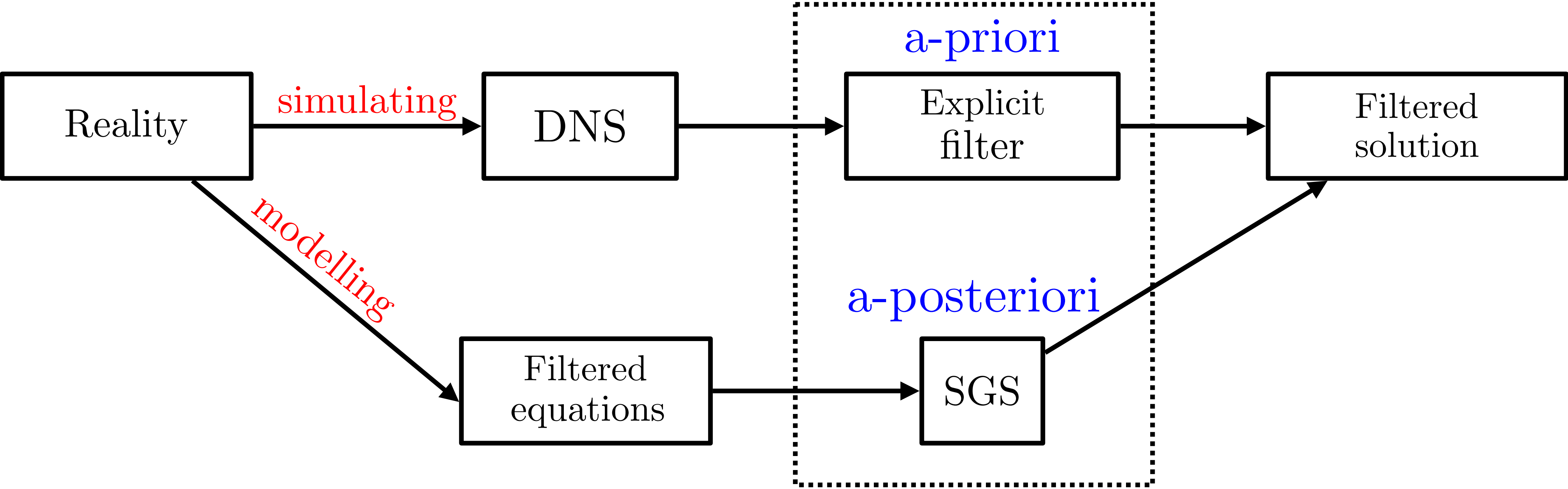
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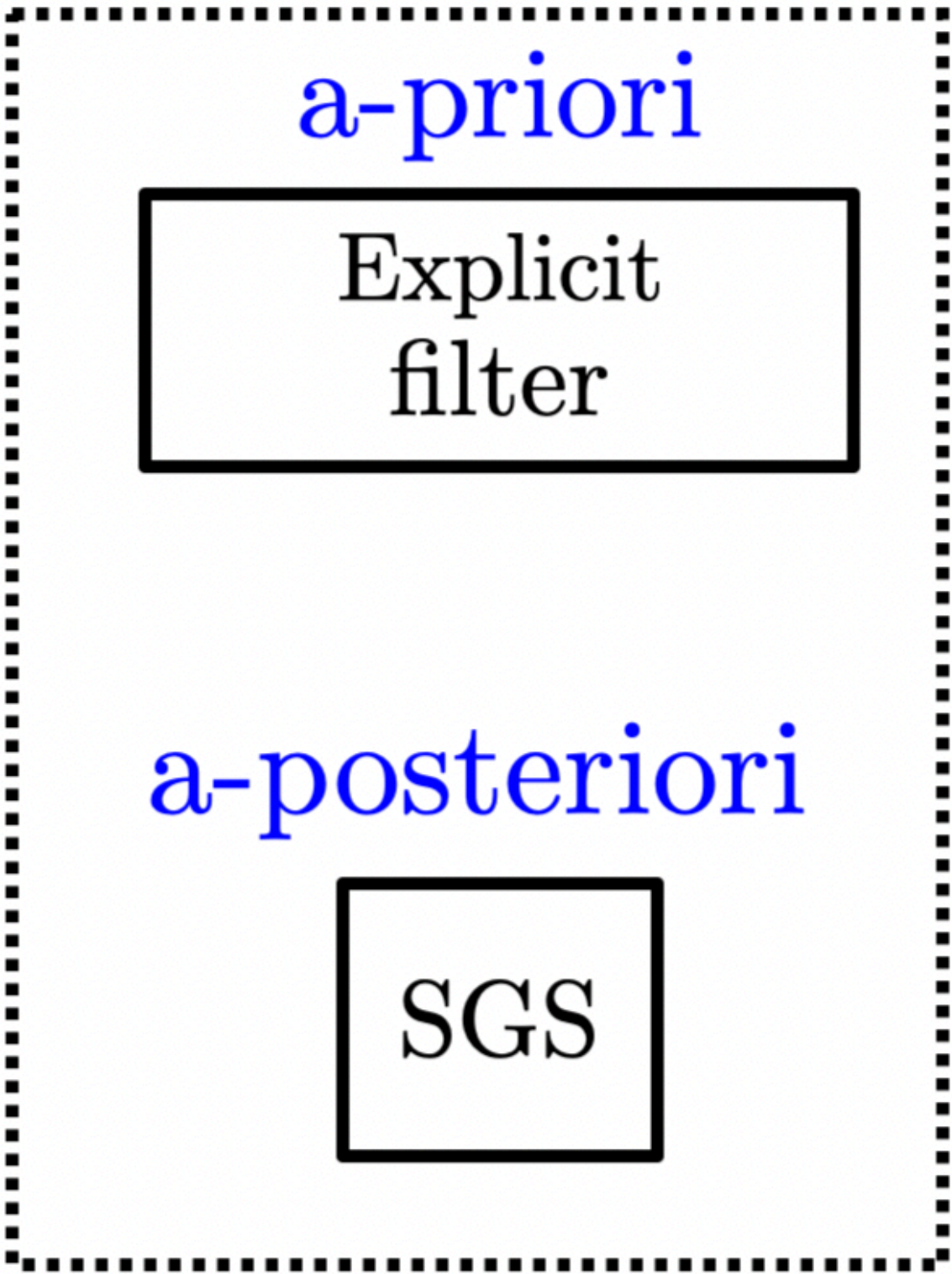
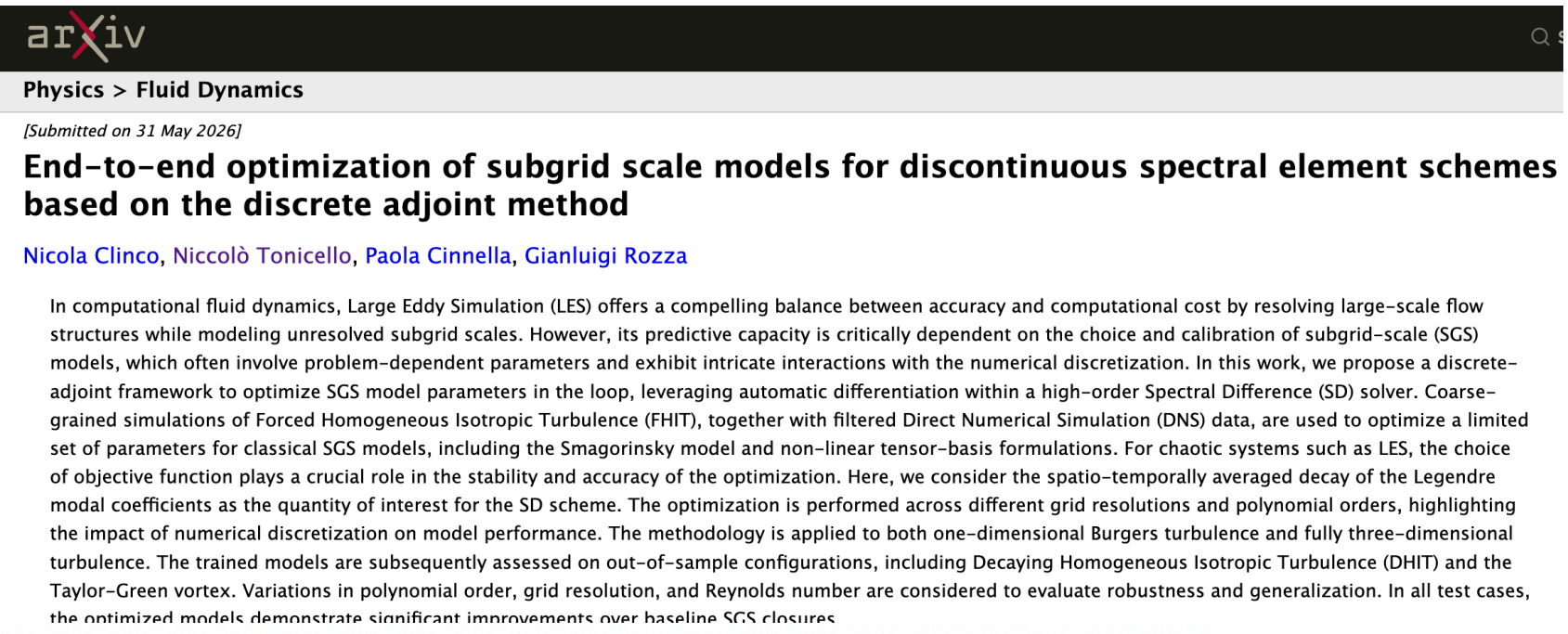
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Simulation Sciences: the role of ML

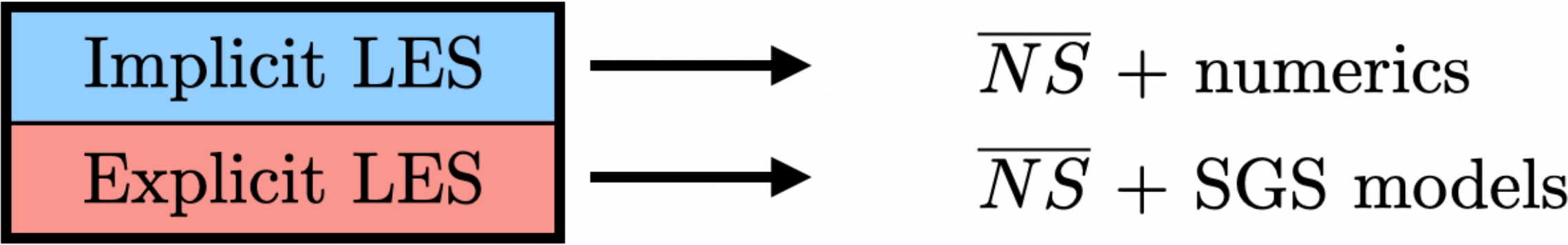
What are we asking to Scientific Machine Learning in the field of Simulation Sciences?

Calibrate models/inverse problems:



$$\overline{\phi}(x, t) = \mathcal{F}(x, \Delta) \star \phi(x, t)$$

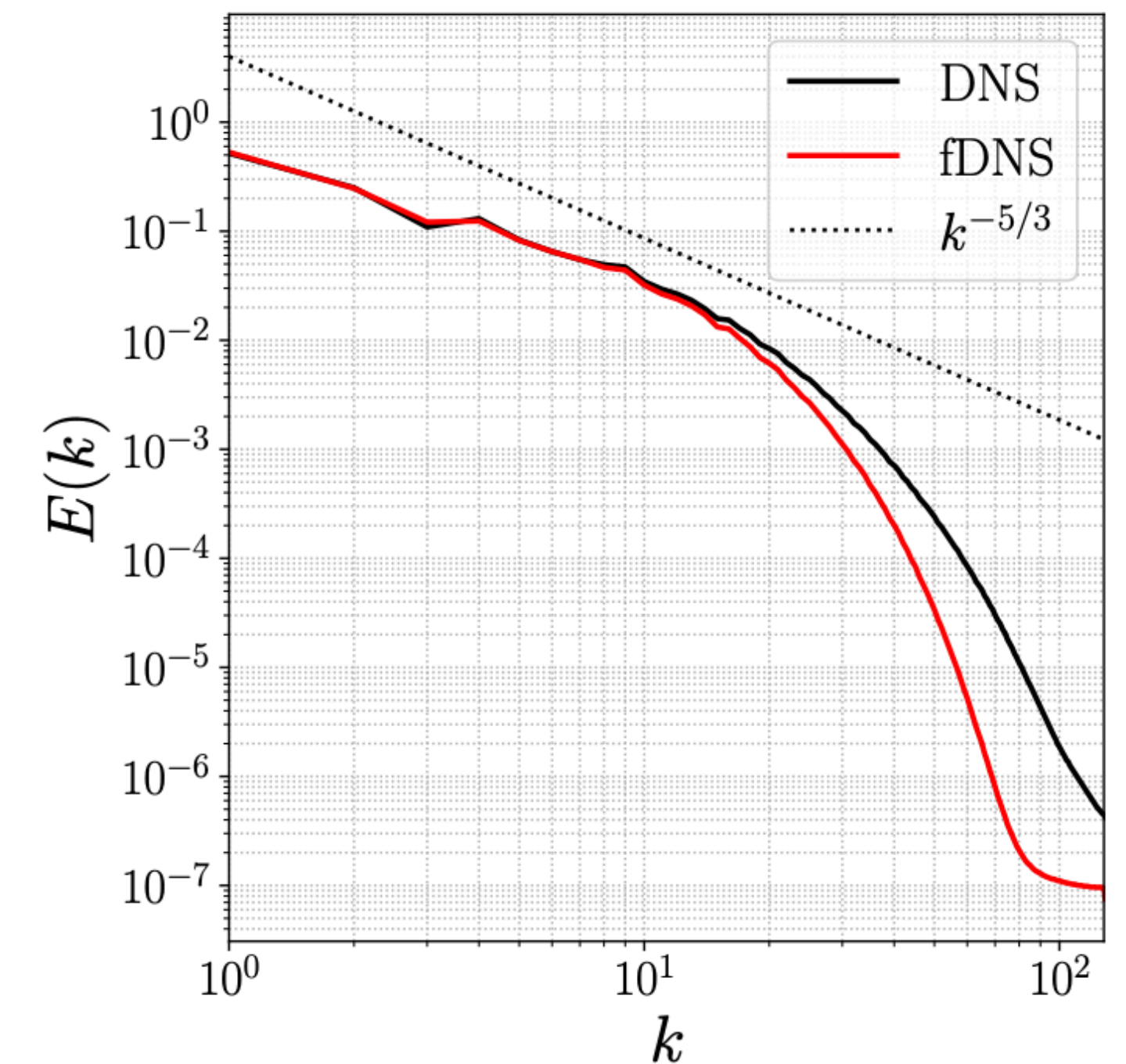
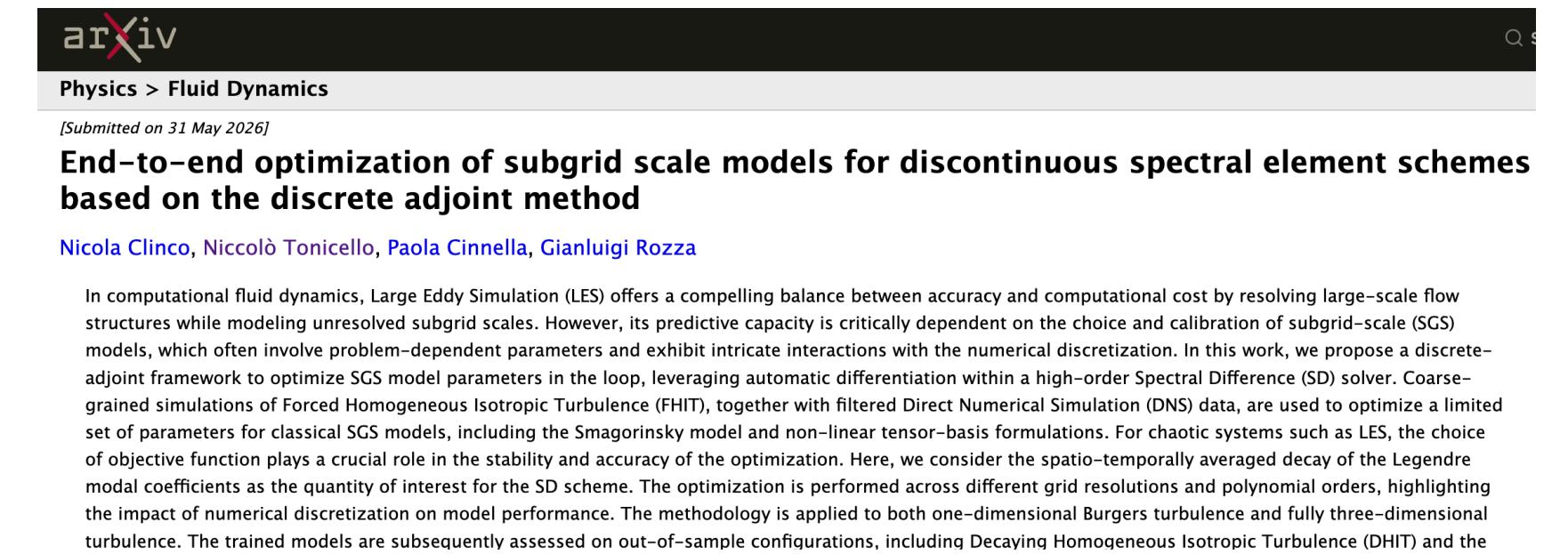
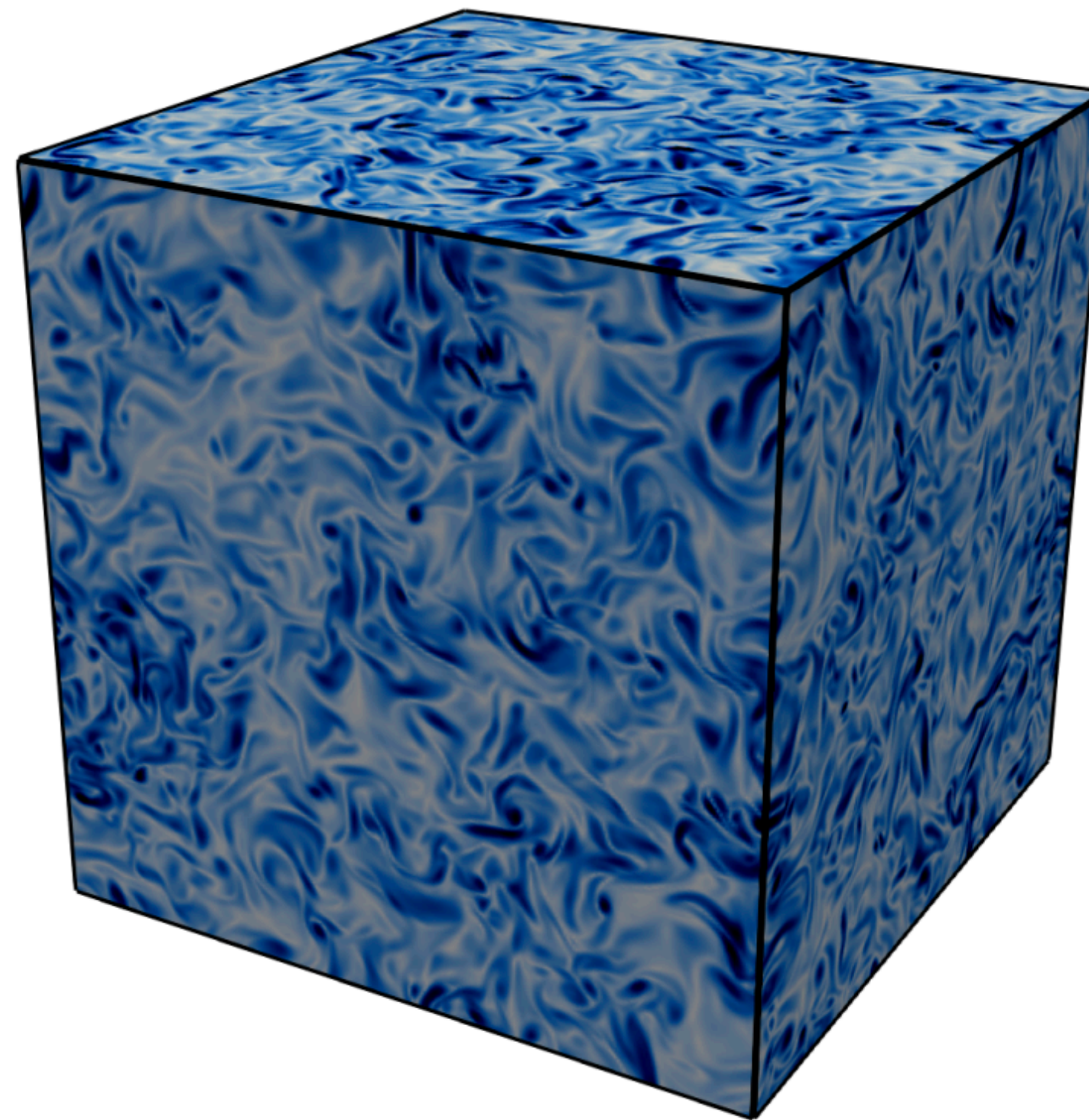
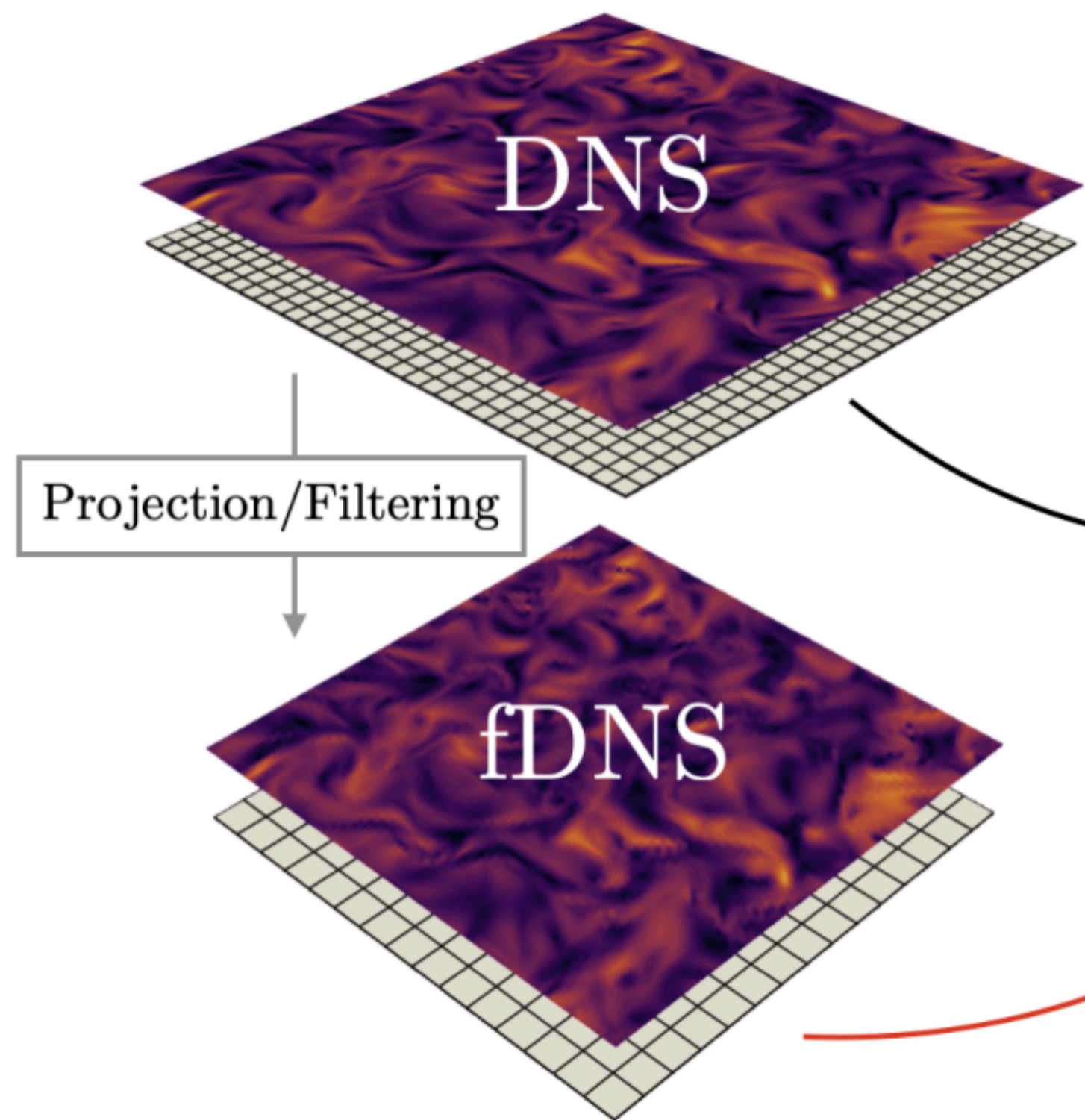
Gaussian?
Spectral cut-off?



Simulation Sciences: the role of ML

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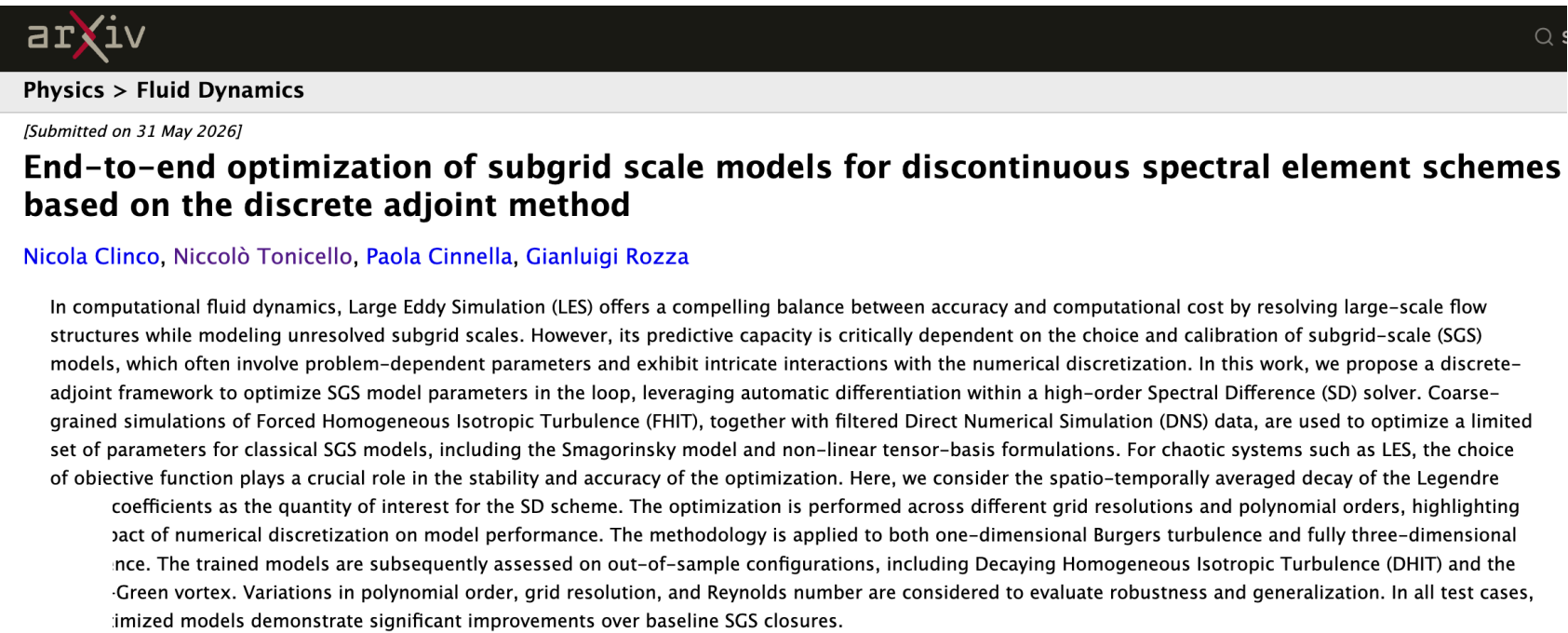
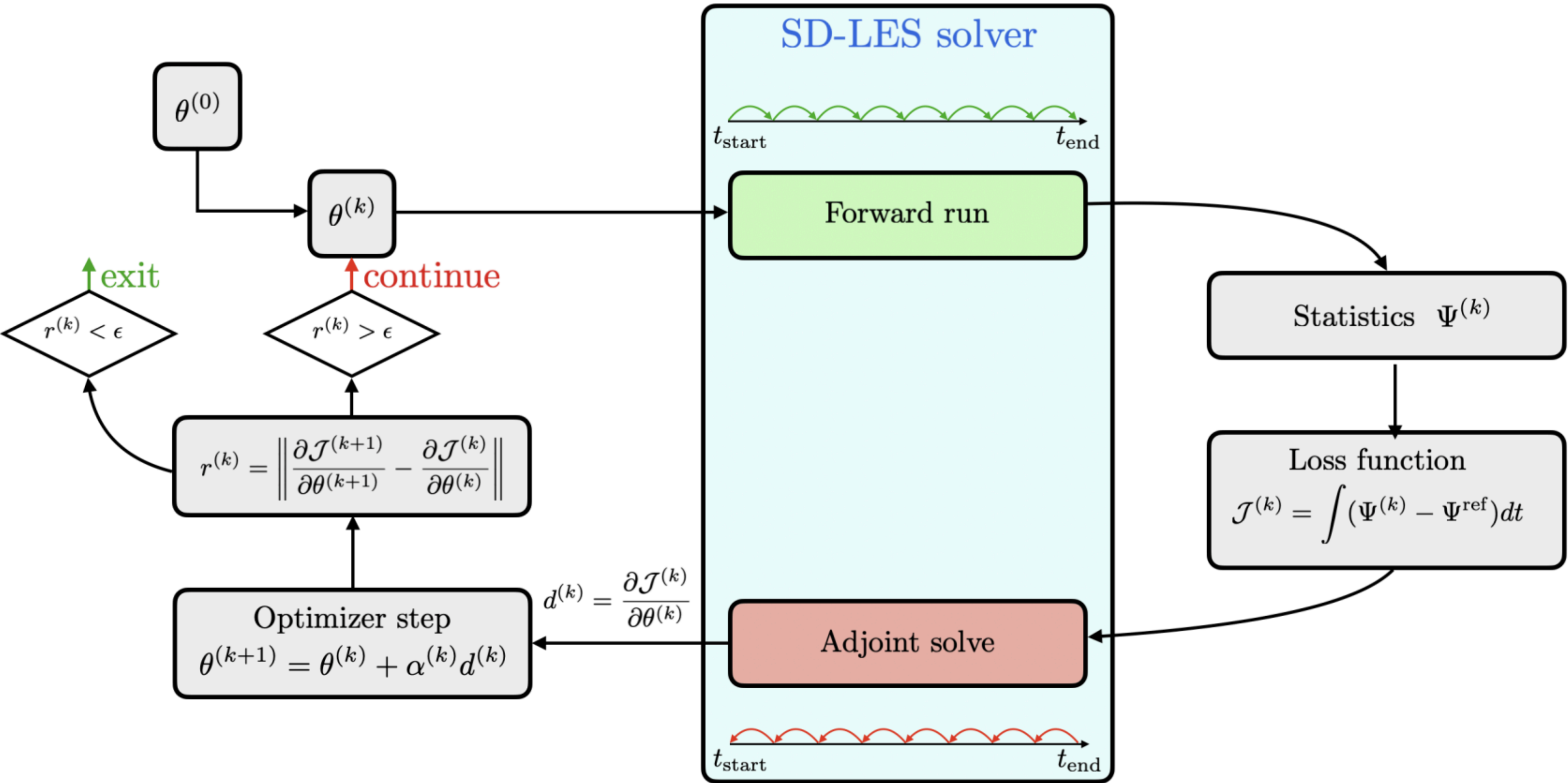
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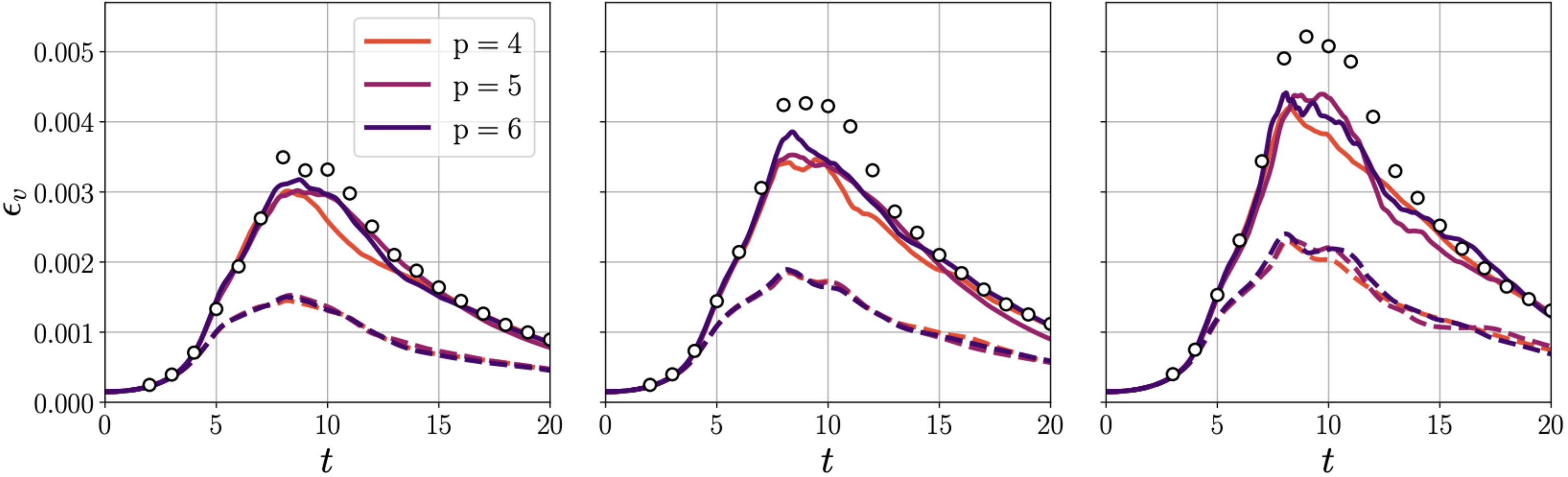


We tackled this problem using the discrete adjoint method.

Simulation Sciences: the role of ML

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arXiv

Physics > Fluid Dynamics

[Submitted on 31 May 2026]

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We were able to optimise the coefficients of classical turbulence models to match the reference for different resolutions, different flow regimes and different numerical details (such as the polynomial order of approximation)